

B cell expression of the enzyme PexRAP, an intermediary in ether lipid biosynthesis, promotes antibody responses and germinal center size

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This study provides **useful** insights into the ways in which germinal center B cell metabolism, particularly lipid metabolism, affects cellular responses. The authors use sophisticated mouse models to **convincingly** demonstrate that ether lipids are relevant for B cell homeostasis and efficient humoral responses. The authors then conducted in vivo as well as in vitro experiments, thereby strengthening their conclusions.

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Abstract

The qualities of antibody (Ab) responses provided by B lymphocytes and their plasma cell (PC) descendants are crucial facets of responses to vaccines and microbes. Metabolic processes and products regulate aspects of B cell proliferation and differentiation into germinal center (GC) and PC states as well as Ab diversification. However, there is little information about lymphoid cell-intrinsic functions of enzymes that mediate ether lipid biosynthesis, including a major class of membrane phospholipids. Imaging mass spectrometry (IMS) results had indicated that concentrations of a number of these phospholipids were substantially enhanced in GC compared to the background average in spleens. However, it was not clear if biosynthesis in B cells was a basis for this finding, or whether such cell-intrinsic biosynthesis contributes to B cell physiology or Ab responses. Ether lipid biosynthesis can involve the enzyme PexRAP, the product of the *Dhrs7b* gene. Using combinations of IMS and immunization experiments in mouse models with inducible *Dhrs7b* loss-of-function, we now show that B lineage-intrinsic expression of PexRAP promotes

the magnitude and affinity maturation of a serological response. Moreover, the data revealed a *Dhrs7b*-dependent increase in ether phospholipids in primary follicles with a more prominent increase in GC. Mechanistically, PexRAP impacted B cell proliferation via enhanced survival associated with controlling levels of ROS and membrane peroxidation. These findings reveal a vital role of this peroxisomal enzyme in B cell homeostasis and the physiology of humoral immunity.

Introduction

The qualities and concentrations of antigen (Ag)-specific antibodies (Ab) are key elements of immunity and the pathophysiology of diverse conditions involving inflammation (1–3). Ab are secreted by cells in the B lymphocyte lineage, but the exact sources are diverse (4–7). Many derive from a subset termed B1 B cells, which are thought to having characteristics of innate responses (4, 5), but progeny of follicular and marginal zone B cells (FoB and MZB, respectively) can differentiate into Ab-secreting plasma cells (PC) after activation and proliferation (6–8). Ab concentrations and affinity for Ag, as well as Ab isotype after class switching in a B cell, determine the efficacy of pathogen clearance (9–12). Importantly, the affinities of serum Ab for target Ag can increase over time after an immune exposure, especially after recurrent encounter(s) (8, 9, 13, 14).

Many sources and forms of Ab can be protective, though in several autoimmune diseases these molecules and the somatic mutations that diversify an initial BCR repertoire drive pathology (15, 16). The micro-anatomic structure termed the germinal center (GC) provides one mechanism to increase spectrum of Ab affinities over time and repeated Ag exposure (17–19). Although T cell help-dependent diversification and PC production can occur independent from GC, results with different forms of vaccination support the practical importance of GC (20, 21). Current evidence holds that most Ig class-switching is executed before GC entry of proliferating B cells (22). GC reactions are initiated after B cells in a primary lymphoid follicle encounter Ag, migrate to interact with activated helper T cells after extensive proliferation (23). GC formation, size, and function depend on multiple factors. These include the efficiency of population growth for B cells in a phase before they enter and take on characteristics of GC B cells, and on homeostatic and differentiative processes while in the secondary follicle [reviewed in (24)]. Once in the GC, B cells cycle between light and dark zones, undergoing iterative cycles of selection. Proliferation and selection end in development as Ab-secreting plasma cells or memory B cells [reviewed in (14, 18, 19)]. In line with a selection process, substantial rates of B cell death have been measured in GC (25). As such, the qualities and quantities of Ab elicited by immunization (or vaccination) depend on B cell proliferation and survival both before and during GC reactions.

Accumulating evidence indicates that metabolic reprogramming and intermediary metabolism play pivotal roles in survival, proliferation, differentiation and function in the pre-immune populations (26, 27), including B cells (28, 29). Among core metabolic processes, several converge on oxidative metabolism fed by glycolytic generation of pyruvate, anaplerotic use of amino acids such as glutamine, and fatty acid oxidation [reviewed in (29, 30)]. Compared to resting populations such as naive B cells, rates of such metabolism in B cells appear to be increased after activation, including in GC B cells [31–36; reviewed in (29)]. Such increases support the generation of substrates for anabolic processes in the dividing cell population and perhaps the energy demands thereof.

Mitochondria have been the focus of investigations exploring molecular mechanisms for oxidative metabolism to generate energy and substrates in lymphocytes during immune responses [reviewed in (37, 38)]. Oxidative phosphorylation in and dynamics of this organelle appear to be important in determining rates of somatic hypermutation that characterize GC B cell biology

(33–35), albeit by a molecular mechanism that has not yet been identified. Among specialized functions of this highly dynamic subcellular organelle, mitochondria use the Krebs TCA cycle to feed an electron transport chain (ETC) and efficiently generate ATP. Along with other sources, mitochondria generate reactive oxygen species (ROS) via the ETC at a rate determined by structural and biochemical features (39, 40). Rates of ROS production, release, and resolution require regulation at multiple levels because these labile species and their reactions with cellular molecules are important for both signaling and cell proliferation but undermine cell function or survival when excessive.

B cells in most GC exhibit inadequacy of oxygen delivery relative to demands (31, 41–43). Of note, genome-wide screening for genes that help cells adapt to such hypoxia revealed that peroxisomes and an ER enzyme involved in synthesizing plasmalogen ether lipids (EL) decrease the death of cells due to insufficient oxygen (44). Using an unbiased lipidomic analysis that combined imaging mass spectrometry (IMS) and techniques for identifying GC in sections of spleen from immunized mice, we had found that local concentrations of a subset of EL were heightened in GC relative to other portions of the spleen (45). Ether phospholipids (EPL) have a fatty acid linked to the glycerol backbone with an alkyl or a vinyl ether bond at the sn-1 position (46, 47). Synthesis of ether phospholipids depends on the generation of precursors in peroxisomes and final processing in the ER (48, 49). However, EL biosynthesis is prominent in the liver, and in theory these species or their precursors might distribute to tissues from a primary site of biosynthesis or from the diet (49–53). Thus, the finding that secondary follicles had greater densities of some - but not all - ether phospholipids left several key questions unanswered – (1) Does the capacity to synthesize and increase ether lipids in B cell follicles – primary or secondary (i.e., GC) - affect antibody responses? (2) Are any of the EL dependent on B lineage-intrinsic metabolism to generate precursors or final products? Moreover, there is debate whether EL or their plasmalogen subset promote cell death or resistance, and whether inactivation of biosynthesis causes defects in establishment or maintenance of hematopoietic cells (46, 53–55).

Peroxisomal Reductase Activating PPAR γ (PexRAP, encoded by the *Dhrs7b* gene) catalyzes the reduction of alkyl-dihydroxyacetonephosphate (DHAP) to 1-alkyl-G3P (46, 48, 56), which then is transferred into endoplasmic reticulum (ER) and further metabolized to generate EL. Recent work notes that this enzyme may substantially function in the ER in addition to the peroxisome (56). Widespread inactivation of *Dhrs7b* in mature mice decreased hepatic production of ether lipid species and reduced the population of erythrocytes and neutrophils (46). This body of work indicated that PexRAP can be essential in some aspects of ether lipid metabolism and may function in hematopoietic cells, but was questioned based on inborn errors of metabolism affecting other aspects of ether lipid biosynthesis (54). Diverse functions of ether lipids in general, and the plasmalogen subset in particular, have been proposed or supported by prior work [reviewed in (57–61)]. The ether linkage may provide a sink for oxygen radicals - thereby acting directly to resolve the reactivity and contribute to ROS homeostasis, although there is debate as to the physiological impact of this chemical property (62). Whether or not this lipid subset influences adaptive immunity or conventional lymphocyte lineages, however, is unclear.

Herein, we tested the hypotheses that (i) the ether lipid profiles in activated B cells depend on their expression of the biosynthetic enzyme PexRAP [alternatively referred to as acyl/alkyl-DHAP reductase (ADHAPR) (56)], and (ii) this enzyme promotes B cell physiology. To do so, we combined conventional lipidomics, Imaging Mass Spectrometry (IMS), and immunization experiments with genetic models in which loss-of-function for *Dhrs7b* is induced in adult mice. We provide evidence that PexRAP impacts B cell proliferation via enhanced survival associated with controlling levels of ROS and membrane peroxidation. The results also indicate that beyond an impact on B cell homeostasis, GC are reduced in its absence, and the magnitude and affinity

maturation of a serological response depend on B cell expression of PexRAP. Taken together, these findings support a function of B cell-intrinsic biosynthesis of ether lipids in B lymphocyte homeostasis, the production of Ab, and in promoting GC reactions.

Results

PexRAP promotes homeostatic maintenance and proliferation of B cells

By use of IMS, we had reported that at least a dozen ether phospholipids - including plasmalogens - were enriched in splenic germinal centers of immunized mice when compared to the rest of the tissue (45). Previous work provided evidence that these lipid molecules were important for the homeostasis of short-lived neutrophils (46). However, this effect was questioned based on hematological data from patients inborn errors of metabolism that affect ether lipid biosynthesis (54). Moreover, the impact of these molecules on adaptive immune cells or functions is not known. Accordingly, we sought to test for effects of an intervention that interferes with a key biosynthetic pathway used for endogenous synthesis of these molecules. To do so, we started with analyses using induced loss-of-function for *Dhrs7b*^{ff} and the widely expressed *Rosa26-CreER*^{T2} transgene (63, 64) used in (46), which reduced hepatic generation of a subset of ether lipids and acts in all hematopoietic cells as well.

We modeled initial experiments on the published work with imaging mass spectrometry (45). Mice were treated with tamoxifen using conditions less intense than a regimen shown to have no effect on pre-immune splenic and B lineage populations (65, 66), immunized with SRBC [akin to the analyses in (31, 36, 41, 45)], and analyzed 7 d after immunization (Fig. 1A). Functional inactivation of *Dhrs7b* in T and B lymphocytes was confirmed by Western blot analysis using anti-PexRAP Ab (Fig. 1B). The frequencies and numbers of CD19⁺ B220⁺ B cells in *Dhrs7b*^{ff}; *Rosa26-CreER*^{T2} - only a small minority of which would have been activated by the immunization - were approximately 0.6 the values for controls (*Dhrs7b*^{+/+} *Rosa26-CreER*^{T2}, i.e., wild-type locus) (Fig. 1C, D). In contrast to B cell populations, TCRβ⁺ CD4⁺ T cells and TCRβ⁺ CD4⁻ CD8 T cells were intact in tamoxifen-treated and SRBC-immunized *Dhrs7b*^{ff}; *Rosa26-CreER*^{T2} mice (Supplemental Fig. 1A-C). These results suggest that *Dhrs7b* is required for fully maintaining the B cell population but indicate that the enzyme is dispensable for the main sets of conventional T cells. In light of the earlier finding that a subset of ether lipids was more prevalent in secondary follicles, we analyzed GC in these samples. Microscopy and flow cytometry analyses after immunofluorescent staining found that less GC B cells were present after immunization of *Dhrs7b*^{ff}; *Rosa26-CreER*^{T2} mice, i.e., less IgD⁻ CD95⁺ GL7⁺ (GC-phenotype) B cells (Fig. 1E-I; Supplemental Fig. 1D). Quantitation of the immune fluorescence micrography with spleens from immunized mice revealed that the numbers of GC per spleen were halved in *Dhrs7b*^{ΔΔ} mice compared with controls (Fig. 1I; Supplemental Fig. 1E). The sizes of those GC that did form were substantially reduced as well (Fig. 1I; Supplemental Fig. 1F). The lower number of B cells found in immunized mice - while a less profound effect than the reduction in GC - tempers the result, and it was possible that the function of helper CD4 T cells was impaired even though numbers were not affected. Consistent with this hypothesis, the frequencies of Tfh and GC-Tfh cells were lower in *Dhrs7b*^{ΔΔ} mice compared with controls (Supplemental Fig. 1G-I). While not addressing whether or not B cells are part of the requirement for PexRAP in promoting GC, these results indicate that *Dhrs7b* is necessary for a normal-sized germinal center response.

To determine if there are B lineage-specific functions of PexRAP, we used a conditionally active Cre transgene that is expressed specifically in mature B cells (67). *Dhrs7b* f/f, *huCD20-CreER*^{T2} mice and *huCD20-CreER*^{T2} were treated with tamoxifen to induce Cre-mediated recombination of the floxed alleles after initial establishment of normal B cell populations (Fig. 2A). Western blot analyses showed an almost complete absence of PexRAP protein from B cells purified from

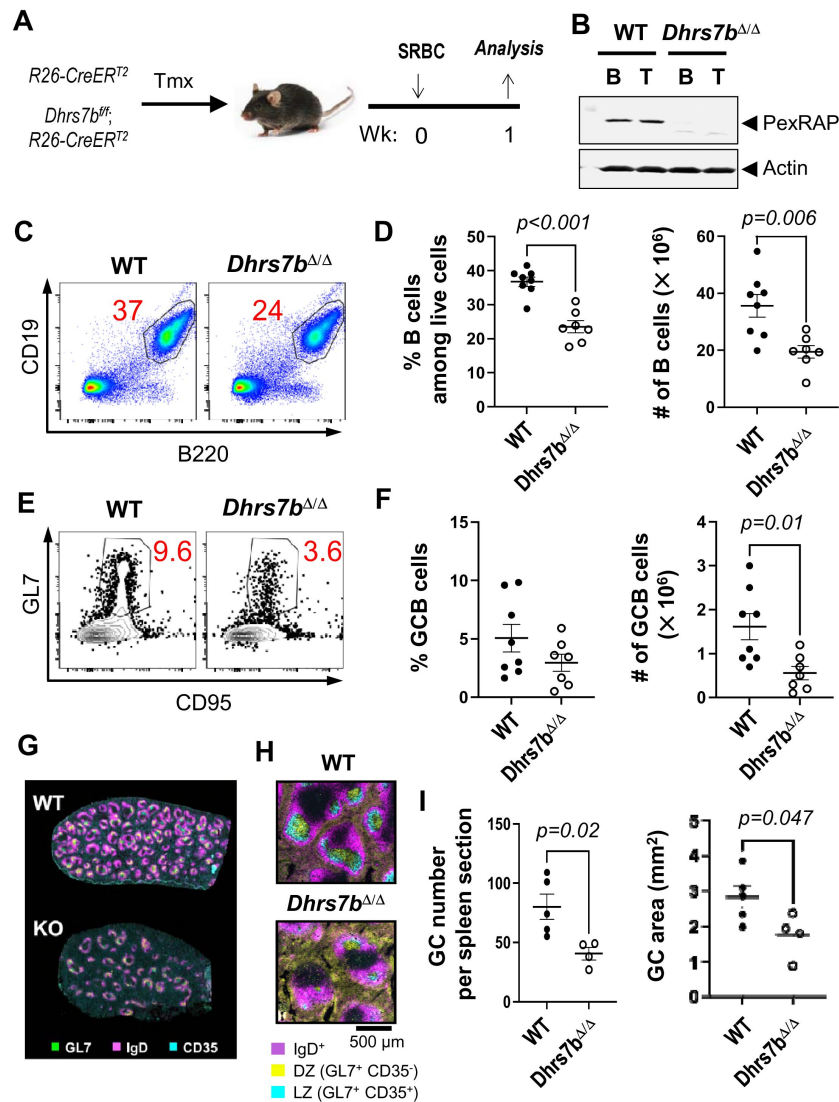


Figure 1.

PexRAP promotes GC response.

(A) Schematic of immunization with SRBC after inactivation of *Dhhrs7b* in adult mice. Mice (*Rosa26-CreERT2*; *Dhhrs7b^{fl/fl}*, or *Rosa26-CreERT2*; *Dhhrs7b^{fl/fl}*) were treated with tamoxifen, immunized with SRBC, and harvested 1 wk after immunization, as described in *Materials and Methods*. (B) Deletion efficiency of *Dhhrs7b* conditional alleles. B and T lymphocytes were isolated from spleens of *Rosa26-CreERT2* or *Dhhrs7b^{fl/fl}; Rosa26-CreERT2* mice after in vivo tamoxifen injections followed by immunization with SRBC. WT and *Dhhrs7b^{Δ/Δ}* were analyzed by immunoblotting with antibodies directed against PexRAP (protein product of *Dhhrs7b*) and actin (internal loading control). (C, D) PexRAP acutely regulates B cell numbers. Representative flow plots of viable splenic B cells (C) and aggregate data for three biologically independent replicate experiments (D) (n = 8 WT and 7 cKO). For data on T cells, see **supplemental data Fig. S1A–C**. (E, F) Effect of PexRAP on GC B cell response. Flow plots of GL7⁺ CD95⁺ GC B cells in the gate for viable IgD-negative (IgD^{neg}), dump-negative B cells (E), and aggregated frequencies and numbers of GC B cells, as indicated, in the three replicate experiments (F). (G, H) PexRAP impact on GC response. After tamoxifen injections, *Dhhrs7b* deficient mice [*Rosa26-CreERT2*; *Dhhrs7b^{fl/fl}* (46)] and WT controls (*Rosa26-CreERT2*) were immunized with SRBC and analyzed 7 d later, as in **Fig. 1A–F**. Data on gating strategy, GC counts per unit area and LZ area are in **Supplemental Fig. 1D–F**, and on Tfh cells in **Supplemental Fig. 1G–I**. (G, H) Shown are representative images from immunofluorescence staining of spleens with the indicated Ab in two independent experiments (5 WT vs 4 *Dhhrs7b* cKO, i.e., *Dhhrs7b^{Δ/Δ}*), showing a low-power overview with many follicles (G) and a representative higher-magnification image to better delineate primary and secondary follicles (H). (I) Quantitation of number (left panel) and size (right panel) of GC in spleen sections, with GL7⁺, IgD^{neg} areas that include both CD35⁺ (LZ) and CD35^{neg} (DZ) areas. Mann-Whitney U test was used to calculate *p* values.

tamoxifen-treated *Dhrs7b^{ff}*; *huCD20-CreER^{T2}* (hereinafter, *Dhrs7b^{Δ/Δ-B}*) mice compared to similarly treated *huCD20-CreER^{T2}* controls (Fig. 2B). The enzyme encoded by this gene catalyzes the synthesis of a lipid precursor to many - but not all - plasmalogens and other ether phospholipids (46–48), and *Dhrs7b* deficiency impacted neutrophil survival (46). When we tested if *Dhrs7b* inactivation affected the steady-state B cell population, the frequencies and total numbers of CD19⁺ B220⁺ B cells in viable lymphocyte gates were about 20% lower in tamoxifen-treated *Dhrs7b^{Δ/Δ-B}* mice compared to tamoxifen-injected, *huCD20-CreER^{T2}*, *Dhrs7b^{+/+}* controls (Fig. 2C). With loss-of-function after production of mature B cells, i.e., inactivation of *Dhrs7b* at ages over 6 wk, we observed balanced decreases in the numbers of multiple subsets within the B lineage without evidence of selectivity among main sub-classes of B cell. Moreover, the surface expression of IgM was comparable in *Dhrs7b^{Δ/Δ}* mice versus WT controls (Supplemental Fig 2, A-F). Thus, the frequencies of MZB and FOB amidst the smaller population of splenic B cells in tamoxifen-treated *Dhrs7b^{ff}*; *huCD20-CreER^{T2}* mice were similar (Supplemental Fig 2B), and frequencies of B1 B cells in the peritoneal cavity also were unaffected (Supplemental Fig 2E). Prior work documents normal frequencies and numbers of developing and mature B cells under harsher conditions of tamoxifen-induced deletion with *CreER^{T2}* (65). De novo B cell production rates are too low to yield the ~20% decrease observed one week after gene disruption, and few splenic B cells are in cell cycle. Accordingly, we infer that after maturation of a B cell, *Dhrs7b* promotes its survival, albeit to a modest degree.

B cell population growth can be impacted by increased cell death and/or decreased proliferation. To test if PexRAP affects proliferation *in vivo*, *Dhrs7b^{Δ/Δ}* B cells were stained with CellTrace Violet (CTV) and adoptively transferred into B cell deficient μ MT recipient mice. Strikingly fewer PexRAP-depleted B cells were recovered (Fig. 2D), and the frequencies of *Dhrs7b^{Δ/Δ}* B cells that were IgD^{neg} (i.e., had been activated) were half those of controls (Fig. 2E; Supplemental Fig. 2G). As compared to control B cells, CTV partitioning analyses suggested that division rates *in vivo* were modestly lower for the *Dhrs7b^{Δ/Δ}* B cells (Fig. 2E, F). Mitogen-stimulated *Dhrs7b^{Δ/Δ}* B cells that proliferated in culture exhibited substantially less robust division: frequencies of viable cells that had undergone ≥ 3 divisions were halved in *Dhrs7b^{Δ/Δ}* B cells) and yielded ~1/3 as large a progeny population (Fig. 2G-I; Supplemental Fig 2H). Collectively, these data indicate that the *Dhrs7b* gene product supports homeostatic maintenance of a quiescent (pre-immune) B cell population *in vivo* and effective proliferation of B cells.

B cell expression of PexRAP is required for achieving normal concentrations of ether phospholipids

As noted, lipidomic analyses that used 2-dimensional image mass-spectrometry (2D-IMS) discovered that local concentrations of a subset of ether lipids are enriched in GC compared with the area outside of GC after immunization (45). To measure the extent to which PexRAP expression within mature B cells can affect their lipid content and composition, including their ether- and plasmalogen phospholipids, LC-MS-MS analyses were performed with B cells directly isolated from spleens as well as those cultured after mitogenic activation. These analyses showed substantial reductions in many ether phospholipids - as well as lysophosphatidylethanolamines (LPE) generated by phospholipase cleavage of the R2 fatty acid sidechain of an ether phospholipid - in PexRAP-deficient B cells (Fig. 3A, B; Supplemental Fig. 3A, B). While beyond the scope of this work, this evidence suggests that generation of potential signaling molecules via phospholipase(s) such as PLA2 may be reduced. Of note, *Dhrs7b* inactivation reduced levels of a subset of plasmalogens in naive B cells (Fig. 3A, B, D; Supplemental Fig 3A) though changes of others were modest or absent in the naive population. A greater impact of PexRAP on quantitative amounts of phospholipids was found in the activated B cells. Several species, skewed towards those with the most detected ions, a number of which were at less than 1/5th the level of non-deleted control B cells (Fig. 3C, D). Phospholipid results observed with naive and activated B cells from tamoxifen-treated *CreER^{T2}*-expressing mice were similar to those of unmanipulated controls (Supplemental Fig. 3C, D). There is no *in vitro* surrogate that faithfully represents GC

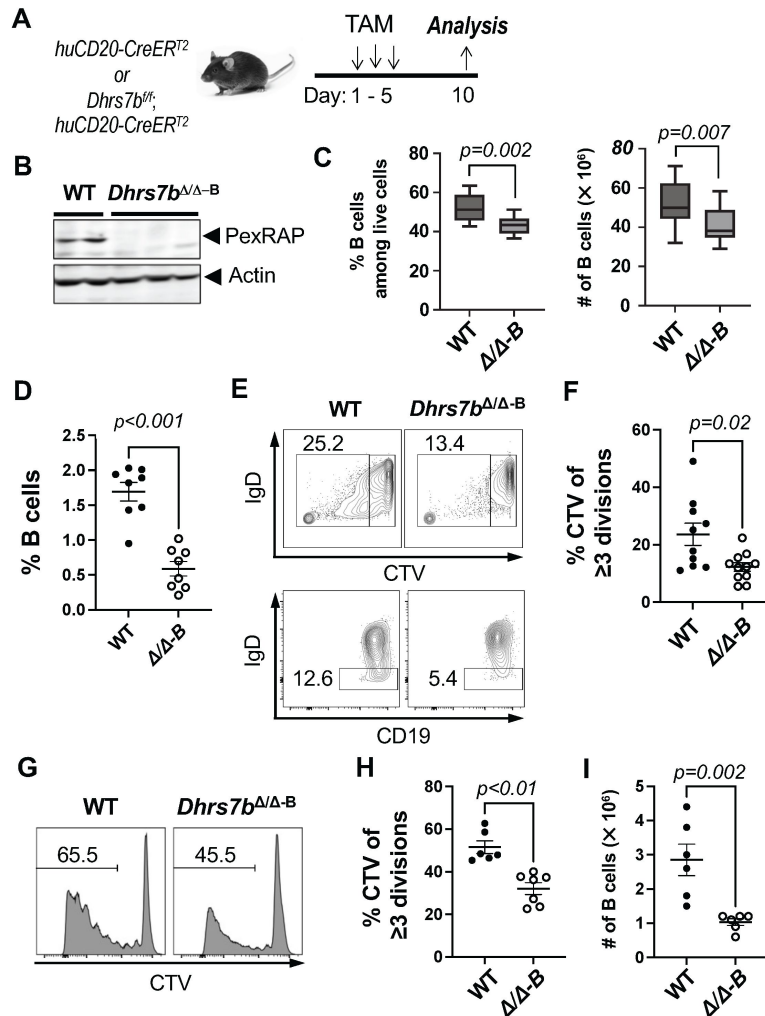


Figure 2.

PexRAP promotes proliferation of B cells.

(A) Schematic of *Dhrs7b* inactivation in mature mice via tamoxifen treatment and a B cell type-specific conditional allele. Mice (*huCD20-CreER^{T2}*; *Dhrs7b^{+/-}*, or *huCD20-CreER^{T2}*; *Dhrs7b^{fl/fl}*) were injected with tamoxifen (d1, 3, 5) and harvested at day 10. (B) Deletion efficiency of conditional *Dhrs7b* alleles. After in vivo tamoxifen injections, B cells were isolated from spleens of *huCD20-CreER^{T2}* or *Dhrs7b^{fl/fl}*; *huCD20-CreER^{T2}* mice, as indicated, and analyzed by immunoblotting as in Fig. 1B. (C) PexRAP and the maintenance of B cells. Shown are the frequencies and the numbers of CD19⁺ B220⁺ B cells among viable lymphocytes in spleen (left and right panels, respectively). Data are pooled from four independent replicate experiments (n = 12 WT and 12 cKO). Shown in box and whisker plots are the means, with whiskers that extend to the minimum and maximum values and boxes that outline upper and lower quartile values with the midline identifying the median. (D-F) PexRAP regulates B cell proliferation in vivo. CTV-labeled B cells were adoptively transferred into μ MT recipient mice and analyzed 4 days thereafter. Shown are the frequencies of B220⁺ CD19⁺ events among splenocytes in the viable cell gate (D), along with representative flow plots of CTV partitioning and surface IgD in B cell gates, with rectangles defining the gating for divided > or ≤ 3 x, and IgD⁺ vs IgD^{neg}. A similar difference was observed in analyzing divided vs undivided. (E) (F) Aggregated frequencies of divided B cells from five independent replicate experiments, as defined in (E) (n = 10 WT and 11 cKO). Additional data on the IgD^{neg} population are in Supplemental Fig. 2G. (G-I) PexRAP promotes B cell proliferation in vitro. After in vivo tamoxifen injections, bead-purified B cells from spleens of CreER^{T2} mice (WT and *Dhrs7b^{Δ/Δ}*, i.e., cKO) were stained with Cell Trace Violet (CTV), activated and cultured 4 days in anti-CD40, BAFF, IL-4, IL-5 and 4-hydroxytamoxifen in three biologically independent experiments totaling 6 WT and 7 cKO mice). (G) Representative flow-cytometric analysis of CTV partitioning, with the gating line denoting multiply divided cells, with inset numbers representing the frequencies of such B cells in each of the two plots shown. (H) Quantified frequencies of divided ≥ 3 times. (I) Numbers of B cells recovered at the end of the cultures.

or the GC B cell, but collectively these data provide strong evidence that the capacity to rapidly produce increased levels of many ether lipid molecules after lymphocyte activation depends on a functional *Dhrs7b* gene in B cells. The data provide evidence that *Dhrs7b* in B cells enhances cell-autonomous biosynthesis of some plasmalogens and other ether lipid species. Especially in resting B cells that are out of cycle, additional uptake or biosynthesis pathways may add to the overall pool (46, 47). Notwithstanding these potential contributions, however, the straightforward inference is that PexRAP catalyzes the generation of ether lipid precursors in amounts crucial for regulating the overall ether phospholipid pools, especially after B cell activation.

Our earlier work (45) reported only ion mode features in negative mode and did not analyze the primary follicle (the vast majority of which are resting B cells) relative to the extrafollicular white pulp, red pulp, and GC. To investigate the requirement for specific expression of PexRAP, we started with AID-GFP mice to enhance spatial localization. These measurements confirmed that a substantial number of both positive and negatively charged ions whose exact masses identified them as plasmalogens were substantially concentrated in GC as compared to the rest of the B cell follicle (Table 1; Supplemental Fig. 4A, B). Accordingly, we tested if the differential enrichment of any ether lipid species in primary or secondary follicles depends on biosynthesis within a specific lymphoid lineage. Mice with conditional mutations were tamoxifen-treated, immunized with SRBC, harvested seven days thereafter, and analyzed by IMS (Fig. 4A, B). With deletion driven by the widely expressed *Rosa26-CreER^{T2}* transgene, as in earlier work (46) and Fig. 1, substantial reductions of the GC enrichment pattern were observed (Fig. 4C). For instance, two representative ions identified in negative ion mode (i.e. *m/z* 752.5545 and *m/z* 776.5556) were again enriched in GC regions of spleens from immunized WT mice. These features barely increased in splenic samples of immunized *Dhrs7b^{Δ/Δ}* (tamoxifen-injected *Dhrs7b^{f/f}*; *Rosa26-CreER^{T2}*) mice (Fig. 4C).

To determine if the levels of particular ether phospholipids in GC regions were affected by PexRAP expression in B cells, *Dhrs7b^{f/f}*; *huCD20-CreER^{T2}* mice and *huCD20-CreER^{T2}* controls were analyzed after tamoxifen injections followed by immunization, and compared to samples from unimmunized mice. IMS using both positive and negative ion modes identified at least eight ions - including *m/z* 752.5545, *m/z* 776.5556, and *m/z* 872.5749 - much more substantially (~2-3-fold) concentrated in GC of immunized control mice (Fig. 4D, E; Table 2) than in the primary follicle of unimmunized mice. Notably, these increased signals in secondary follicles (GC) were reduced or almost completely eliminated in GCs of mice immunized after B cell-specific PexRAP depletion (*Dhrs7b^{Δ/Δ-B}*) (Fig. 4D, E; Table 2; Supplemental Fig. 4C). Collectively, these results indicate that *Dhrs7b* gene expression in activated B cells regulates the spectrum of ether lipids in the micro-anatomic locale of a secondary follicle. Interestingly, immunization also led to modest increases in the levels of some ether and conventional phospholipids in the B cell-rich primary follicle regions (Table 2). This finding extends the previous report (45), which focused on the relationship between the AID-GFP^{hi} region (i.e., GC) and lipid features. Moreover, the absence of PexRAP blunted most of the immunization-induced increases observed in the primary follicles. These findings and those of Fig. 3 suggest that there are secondary consequences of *Dhrs7b* inactivation (e.g., some di-acyl phospholipids are affected), but provide direct evidence that the enhancement of many ether phospholipid signals in the GC depends on PexRAP in B cells.

B cell-intrinsic role of *Dhrs7b* in Ab affinity and quantity

As key components of adaptive immunity, progeny of an activated B cell can differentiate into Ab-secreting plasma cells (PC) or into GC B cells. Over the course of an immune response, the GC reaction diversifies the affinity for and breadth of antigen recognized by the original BCR, and improves properties of memory [(14, 17–19, 68)]. PC that develop after a second encounter with antigen increase the amounts of high-affinity Ab. The higher quality and quantity of Ag-specific Abs from plasma cells are integral to humoral immune responses (68). An

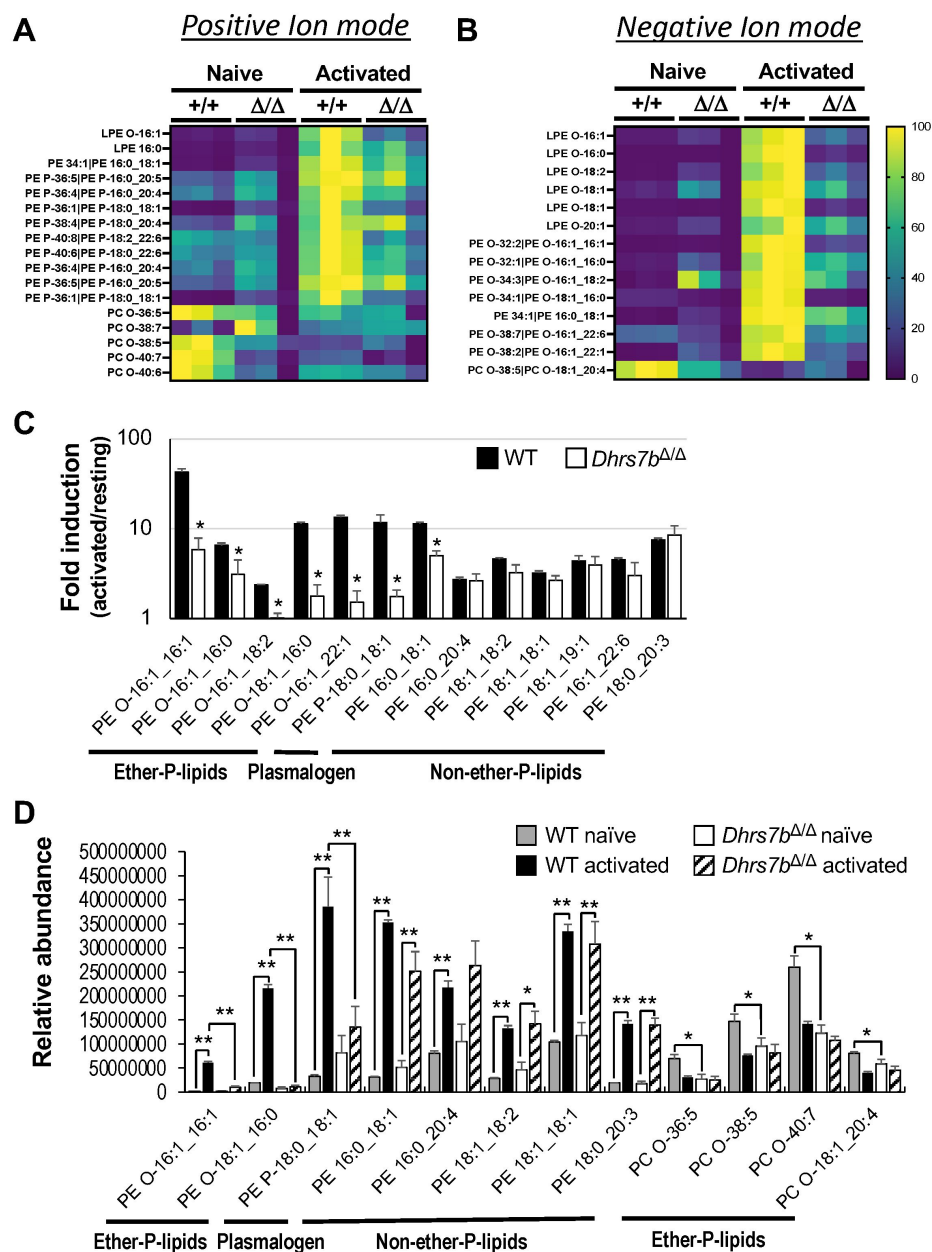


Figure 3.

B cell expression of PexRAP is essential for normal concentrations of most ether phospholipids.

Splenic B cells ($\text{CreER}^{\text{T}2+}$ or $\text{CreER}^{\text{T}2+}$, $Dhrs7b^{\Delta/\Delta}$) from tamoxifen-injected mice were analyzed by LC-MS-MS either directly ex vivo ("Naive") or after activation and culture (48 h) ("Activated") as in [Fig. 2](#). Shown are z-scored heat maps for the relative abundance of subsets of ether and plasmalogen phospholipids and lysophospholipids identified by exact mass and secondary fragmentation in (A) positive and/or (B) negative ion modes. PE, phosphatidylethanolamine; PC, phosphatidylcholine; LPE, lysophosphatidylethanolamine. More species are displayed in [Supplemental Fig. 2](#). (C) Quantitated peak areas for activation-induced increases ('fold-induction' ratios of activated/resting, plotted on $\log_{10}$ scale) for the indicated ether, plasmalogen, and diacyl (non-ether) phospholipids in the PexRAP-sufficient and -depleted B cells. * denotes species for which $P < 0.05$ for the effect of $Dhrs7b$ inactivation. (D) Shown are the raw values of mean peak areas for the indicated species in the freshly purified and the activated B cells of the indicated genotypes, as indicated. (Three independent samples from individual mice of each genotype were analyzed.) * $p < 0.05$, ** $p < 0.01$ by unpaired Student's t-test. Additional data including controls comparing B cells of unmanipulated B6 mice to those expressing $\text{CreER}^{\text{T}2}$ and injected with tamoxifen are in [Supplemental Fig. 3](#).

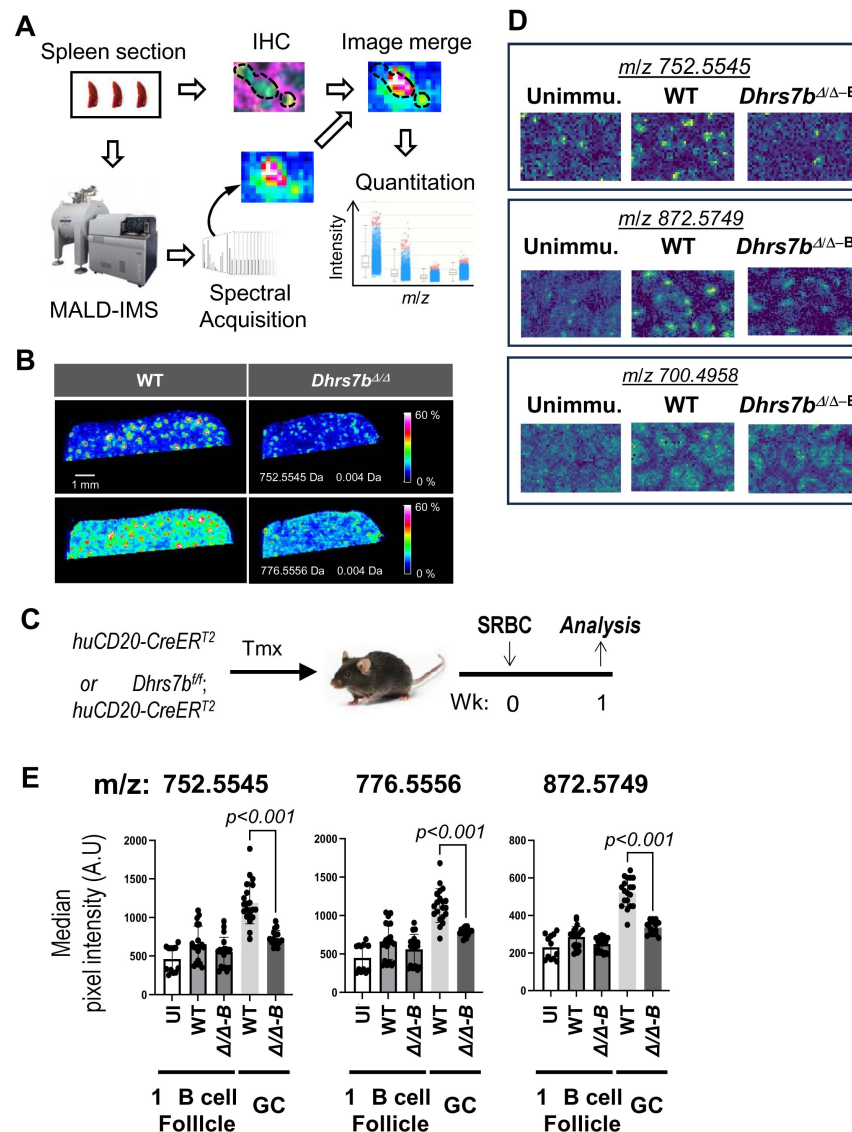


Figure 4.

PexRAP is essential for normal concentrations and distributions of some ether phospholipids in splenic follicles.

(A) Schematic diagram illustrating the 2D-IMS analysis work flow, image merging and quantitation. Spleens harvested 1 wk after immunization with SRBC were used to generate serial tissue sections followed by immunofluorescence (IF) staining of one section and IMS analysis with the adjacent one. IF and IMS images were aligned to map ion intensity distributions to microanatomic regions (B cell follicle and GC). The intensities of specific ions on B cell follicles and GC regions were quantitated as described in *Materials and Methods*. (B) Identification of ether lipid species localizing to lymphoid follicles. Representative ion images of two ions [m/z 752.5545, and m/z 776.5556] with spleens from immunized mice (WT and $Dhhrs7b^{\Delta/\Delta}$) as shown in **Fig. 1A**. (C) Schematic of immunization for IMS analyses of B cell-specific PexRAP loss. Mice of the indicated genotypes ($huCD20-CreER^{T2} \pm Dhhrs7b^{f/f}$) were treated with tamoxifen, immunized with SRBC, and harvested 1 wk after immunization. (D) Identification of ether lipid species localizing to lymphoid follicles. Representative ion images of three ions [m/z 752.5545, m/z 872.5749, and m/z 700.4958] in mass spectrometry imaging of spleens from WT and $Dhhrs7b^{\Delta/\Delta-B}$ mice. Immunofluorescent images at higher magnification, delineating LZ and DZ marked by CD35 staining, along with quantification of sizes of GC and their LZ and DZ, are shown in **Supplemental Fig. 5A, B**. (E) Shown in the bar graphs are the mean ($\pm$ SEM) ion intensities in primary lymphoid follicles (B cell zones) and GC from spleens of WT and $Dhhrs7b^{\Delta/\Delta-B}$ mice, immunized or not ("UI") as indicated. Median intensity of each ion was obtained from 3 follicles / spleen and 3 GC / spleen from WT and $Dhhrs7b^{\Delta/\Delta-B}$ mice (three biological replications comprising 7 WT and 6 cKO spleens). P values were calculated by Mann-Whitney U test.

Negative Ions			
m/z ^a Lipid ID	UI	1° B cell Follicle	GC
716.5216 PE(18:0_16:1)	618 ± 22 ^b	2400 ± 83 ^c	3674 ± 154 ^d
752.5545 PE(O-18:0_20:4)	301 ± 14	1739 ± 33 ^c	1960 ± 52 ^d
760.5056 PE(16:0_22:6)	203 ± 4	1816 ± 30 ^c	2031 ± 23 ^d
772.5242 PE(P-18:1_22:6)	234 ± 5	2334 ± 47 ^c	2578 ± 63 ^d
776.5556 PE(O-18:0_22:6)	277 ± 9	1638 ± 51 ^c	2074 ± 90 ^d
869.5526 PI(O-18:0_20:5)	242 ± 7	1611 ± 24 ^c	1789 ± 15 ^d
872.576 PC(18:0_24:0)	171 ± 6	1167 ± 16 ^c	1369 ± 18 ^d
888.5612 PS(22:1_22:6)	506 ± 13	2684 ± 54 ^c	3386 ± 72 ^d

Positive Ions			
m/z ^a Lipid ID	UI	1° B cell Follicle	GC
739.4587 PG(12:0_20:3)	3042 ± 102	3733 ± 116 ^c	4778 ± 274 ^d
740.465 PC(16:0_15:1)	1310 ± 13	1478 ± 66 ^c	1933 ± 142 ^d
784.5564 PC(10:0_24:0)	2840 ± 24	7956 ± 147 ^c	9856 ± 270 ^d
798.5347 PC(10:0_25:0)	9937 ± 269	27589 ± 574 ^c	34733 ± 1644 ^d
799.5375 PG(18:0_18:1)	5665 ± 65	12078 ± 282 ^c	15311 ± 790 ^d

Table 1.

Immunization-induced increases of selected lipids in GC

Negative Ions					
m/z ^a Lipid ID	UI	WT follicle	PexRap cKO follicle	WT GC	PexRap cKO GC
436.2801 LPE (O-16:1)	315 ± 4	369 ± 7	374 ± 11	610 ± 20 ^e	475 ± 19
716.5216 PE (18:0_16:1)	754 ± 43 ^b	1608 ± 182 ^c	1263 ± 112	2729 ± 247 ^e	1770 ± 137 ^f
746.5098 PE (O-16:1_22:6)	622 ± 19	672 ± 35	609 ± 23	1066 ± 39 ^e	836 ± 39
752.5545 PE (O-18:0_20:4)	459 ± 49	657 ± 51 ^c	548 ± 46 ^d	1191 ± 61 ^e	734 ± 25 ^f
760.5056 PE (16:0_22:6)	229 ± 8	386 ± 25 ^c	312 ± 16 ^d	569 ± 27 ^e	408 ± 15 ^f
772.5242 PE (P-18:1_22:6)	388 ± 47	617 ± 46 ^c	499 ± 33 ^d	918 ± 21 ^e	646 ± 9 ^f
776.5556 PE (O-18:0_22:6)	449 ± 53	661 ± 50 ^c	562 ± 46	1130 ± 49 ^e	789 ± 13 ^f
869.5526 PI (O-18:0_20:5)	509 ± 81	641 ± 50 ^c	494 ± 40 ^d	1014 ± 21 ^e	690 ± 12 ^f
872.576 PC (18:0_24:0)	230 ± 19	285 ± 12 ^c	247 ± 9 ^d	522 ± 17 ^e	336 ± 9 ^f
888.5612 PS (22:1_22:6)	1423 ± 277	2051 ± 199 ^c	1904 ± 199	3260 ± 156 ^e	2629 ± 80 ^f

Positive Ions					
m/z ^a Lipid ID	UI	WT follicle	PexRap cKO follicle	WT GC	PexRap cKO GC
730.5767 PE (P-18:0_18:1)	91 ± 3	97 ± 3	85 ± 3	130 ± 7 ^e	102 ± 4
739.4587 PG (12:0_20:3)	3042 ± 102	3094 ± 40	2944 ± 31 ^d	3506 ± 140 ^e	3011 ± 46 ^f
784.5564 PC (10:0_24:0)	2840 ± 24	2917 ± 74	2617 ± 46 ^d	3522 ± 102 ^e	2561 ± 42 ^f
798.5347 PC (10:0_25:0)	9937 ± 269	10928 ± 214 ^c	11372 ± 155 ^d	15378 ± 116 ^e	14694 ± 186 ^f
801.5509 PG (18:0_18:0)	1540 ± 37	1591 ± 46	1633 ± 26	1444 ± 27 ^e	1533 ± 22 ^f
847.5361 PI (15:0_18:0)	1277 ± 24	1339 ± 54	1217 ± 35	1150 ± 46 ^e	1044 ± 44
874.5623 PC (20:5_22:6)	754 ± 39	688 ± 48	675 ± 25	469 ± 27 ^e	556 ± 37

Table 2.

Impact of PexRAP on relative quantities of selected lipid species in primary and secondary follicles (GC).

increasing body of evidence indicates that metabolic reprogramming and intermediary metabolites modulate immune cell differentiation and function (26–36). The observed B cell-intrinsic function of *Dhrs7b* in the accumulation of many ether lipid species in both primary and secondary follicles prompted us to test the effect of B cell type-restricted PexRAP depletion on GC responses and Ag-specific Ab production. Tamoxifen-injected *Dhrs7b^{fl/fl}*; *huCD20-CreER^{T2}* and control mice were immunized with NP-ovalbumin (NP-OVA), then boosted with NP-OVA to elicit affinity-matured Ab, and harvested 1 week thereafter (Fig. 5A). Although PexRAP-deficient B cell numbers were reduced less than 20% a week after starting gene inactivation prior to immunization (i.e., were over 0.8-fold those of controls) (Fig. 2A–C), frequencies and numbers of IgD^{neg} B cells were halved in *Dhrs7b^{Δ/Δ-B}* mice at harvest (4 wk after completion of the induced deletion) (Fig. 5B, C). Moreover, the GL7⁺ CD95⁺ fraction of the B cell population in the dump / IgD^{neg} gate was substantially reduced (Fig. 5D), such that numbers of GL7⁺ CD95⁺ GC B cells were dramatically decreased in *Dhrs7b^{Δ/Δ-B}* mice (Fig. 5E). Consistent with these findings, GC in *Dhrs7b^{Δ/Δ-B}* mice were reduced after SRBC immunization (Fig. 5F, G; Supplemental Fig. 5A, B), and the numbers of centrocytes and centroblasts were dramatically lower in *Dhrs7b^{Δ/Δ-B}* mice compared with WT controls (Fig. 5H). LZ B cells (CD86⁺ CXCR4^{lo}) trended toward being at lower prevalence in *Dhrs7b^{Δ/Δ-B}* mice, whereas their DZ counterparts (CD86^{neg} CXCR4⁺) were comparable to wildtype controls, resulting in a modest decrease in the LZ/DZ ratio when B cells lacked PexRAP (Supplemental Fig. 5C). The frequencies of CD138⁺ GL7⁺ IgD[−] early plasmablasts were comparable between *Dhrs7b^{Δ/Δ-B}* mice and control mice (Supplemental Fig. 5D). Although the loss-of-function was B cell specific, the frequencies of Tfh and GC-Tfh cells were decreased in *Dhrs7b^{Δ/Δ-B}* mice compared with control mice (Supplemental Fig. 5E). We infer that support for the Tfh populations was reduced due to impact(s) on GC B cell numbers and / or function.

Ab class-switch recombination (CSR) and affinity maturation can occur through extrafollicular responses. However, the GC reaction significantly increases Ab diversification and high-affinity Ab production, in part later in a primary response but also upon secondary exposure to antigens (6, 8, 9, 17–19). ELISA performed with sera at 3 wk post-immunization (Fig. 6A–D), just before the boost, showed that B cell-specific depletion of PexRAP prior to immunization reduced NP-specific IgM and the switched isotype IgG1 (Fig. 6A, C). Of note, the capacity to generate high-affinity Ab, detected with low valency NP₂, was even more severely undermined than the overall response - especially for IgG1 (Fig. 6B, D). A week after a second immunization - an interval similar to that which followed immunization with SRBC - Ab-secreting cells (ASCs) in the spleen (Fig. 6E) and circulating anti-NP IgM concentrations were diminished in *Dhrs7b^{Δ/Δ-B}* mice compared with controls (*huCD20-CreER^{T2}*, *Dhrs7b^{+/+}* treated with tamoxifen in parallel to the mice with induced loss of PexRAP) (Fig. 6F). The ratio of high-affinity (NP₂-binding) to all-affinity (NP₂₀) IgM Ab also was substantially lower in *Dhrs7b^{Δ/Δ-B}* mice (Fig. 6G), as were the serum levels of high-affinity IgG1 and IgG2c, class-switched isotypes (Supplemental Fig. 5F, G). These data reinforce and extend the conclusion that the expression of PexRAP in mature B cells is important for their physiology and function.

Consistent with *in vivo* results finding reduced Ag-specific ASCs in *Dhrs7b^{Δ/Δ-B}* mice compared with controls (Fig. 6E), the attenuated population of PexRAP-deficient B lineage cells recovered after transfer into recipient mice (Fig. 2D) yielded lower frequencies of CD138⁺ progeny (Fig. 6H, I) along with evidence of reduced activation [i.e., lower frequencies of IgD[−] progeny (Supplemental Fig. 2G)]. Mitogen-stimulated *Dhrs7b^{Δ/Δ}* B cells also yielded lower frequencies of CD138⁺ cells *in vitro*, but the division-specific frequencies of CD138⁺ cells showed only modest decreases in *Dhrs7b^{Δ/Δ}* B cells (Supplemental Fig. 6A, B). Thus, the reduction of CD138⁺ cell differentiation appears to be due mostly to its dependence on survival to a sufficient division count.

Dhrs7b inactivation was initiated prior to immunization in the preceding experiments. Therefore, the impact of PexRAP deficiency on GC and the Ab response in such a setting might be due exclusively to impairment of B cells, e.g., their reduced population expansion, prior to their entry into GC. Alternatively, the effects could also in part involve a requirement for PexRAP-dependent

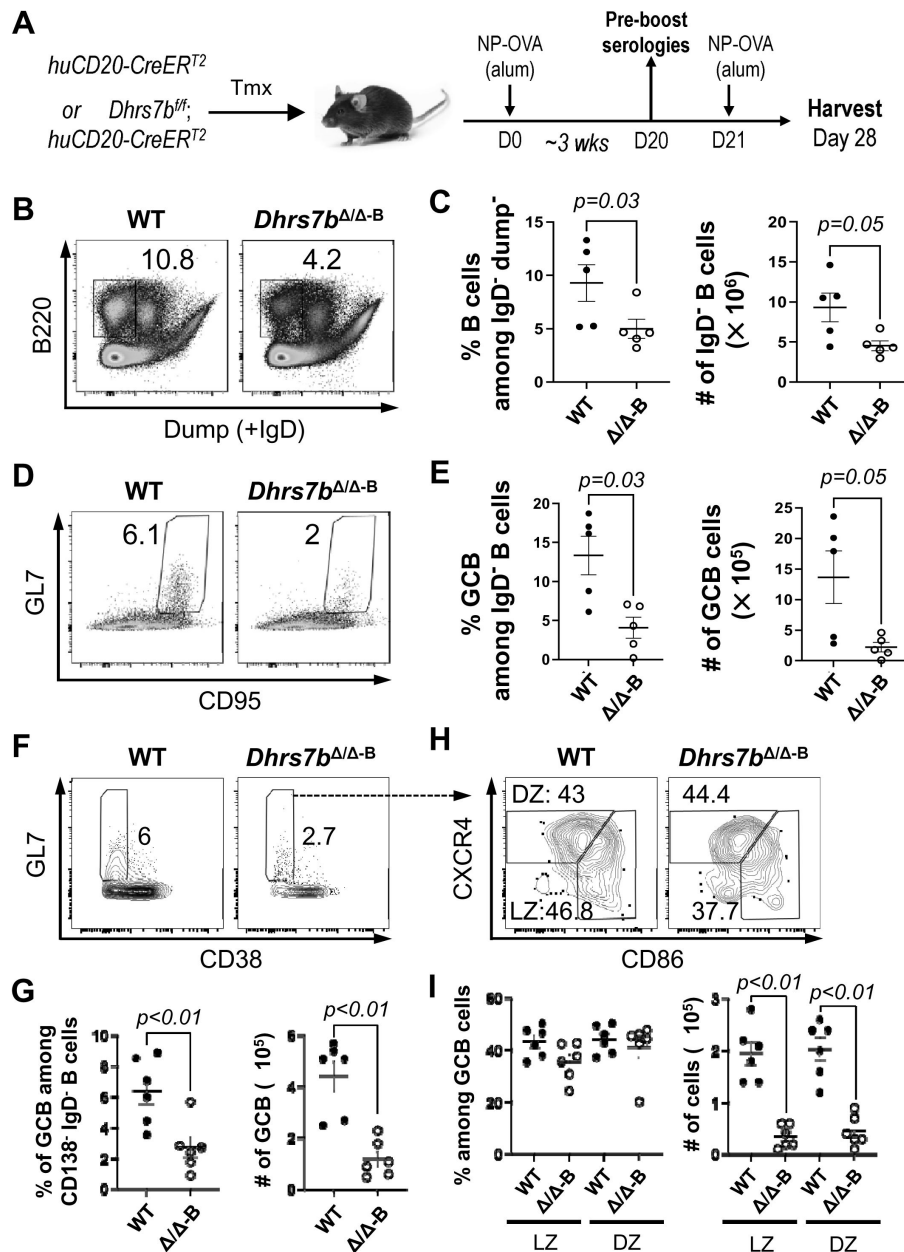


Figure 5.

B cell intrinsic role of PexRAP in GC response.

(A) Schematic of immunization with NP-OVA in alum after inactivation of *Dhrs7b* in B lineage cells. Tamoxifen-treated WT (*huCD20-CreERT2*; *Dhrs7b*^{+/+}) or *Dhrs7b*^{Δ/Δ-B} mice (*huCD20-CreERT2*; *Dhrs7b*^{fl/fl}) were immunized with NP-OVA, with sera collected 3 wk thereafter ("1^o response"), followed by boosting with NP-OVA and harvest 1 wk after the 2nd immunization. (B, C) Representative flow plots of splenic IgD^{neg} B cells (B) at the time of harvest (1 wk after 2nd immunization), and aggregate data (C) for two replicate experiments (n = 5 WT and 5 cKO). The cocktail of reagents for the dump channel included anti-IgD. (D, E) B cell-intrinsic function of PexRAP in GC B cell response. Representative flow plots of GL7⁺ CD95⁺ GC B cells among viable IgD^{neg} B cells (D), and aggregated frequencies and numbers of GC B cells for two replicate experiments (E). Mann-Whitney U test was used to calculate p values. (F, G) WT (*huCD20-CreERT2*; *Dhrs7b*^{+/+}) or *Dhrs7b*^{Δ/Δ-B} mice (*huCD20-CreERT2*; *Dhrs7b*^{fl/fl}) were injected with tamoxifen and immunized with SRBC as in Fig 4A. Shown are the representative flow plots of GL7⁺ CD38⁺ GC B cells among IgD⁻ CD138⁻ viable B cells (F), and aggregate data (G) for two replicate experiments (n = 6 WT and 6 cKO). (H) PexRAP is dispensable for the balance of LZ and DZ B cells. The graph shows the mean (± SEM) frequencies (left) and the numbers (right) of CD86⁺ CXCR4⁺ LZ B cells and CD86^{neg} CXCR4⁺ DZ B cells among IgD^{neg} CD38^{neg} CD138^{neg} GL7⁺ GC cells. Mann-Whitney U test was used to calculate p values.

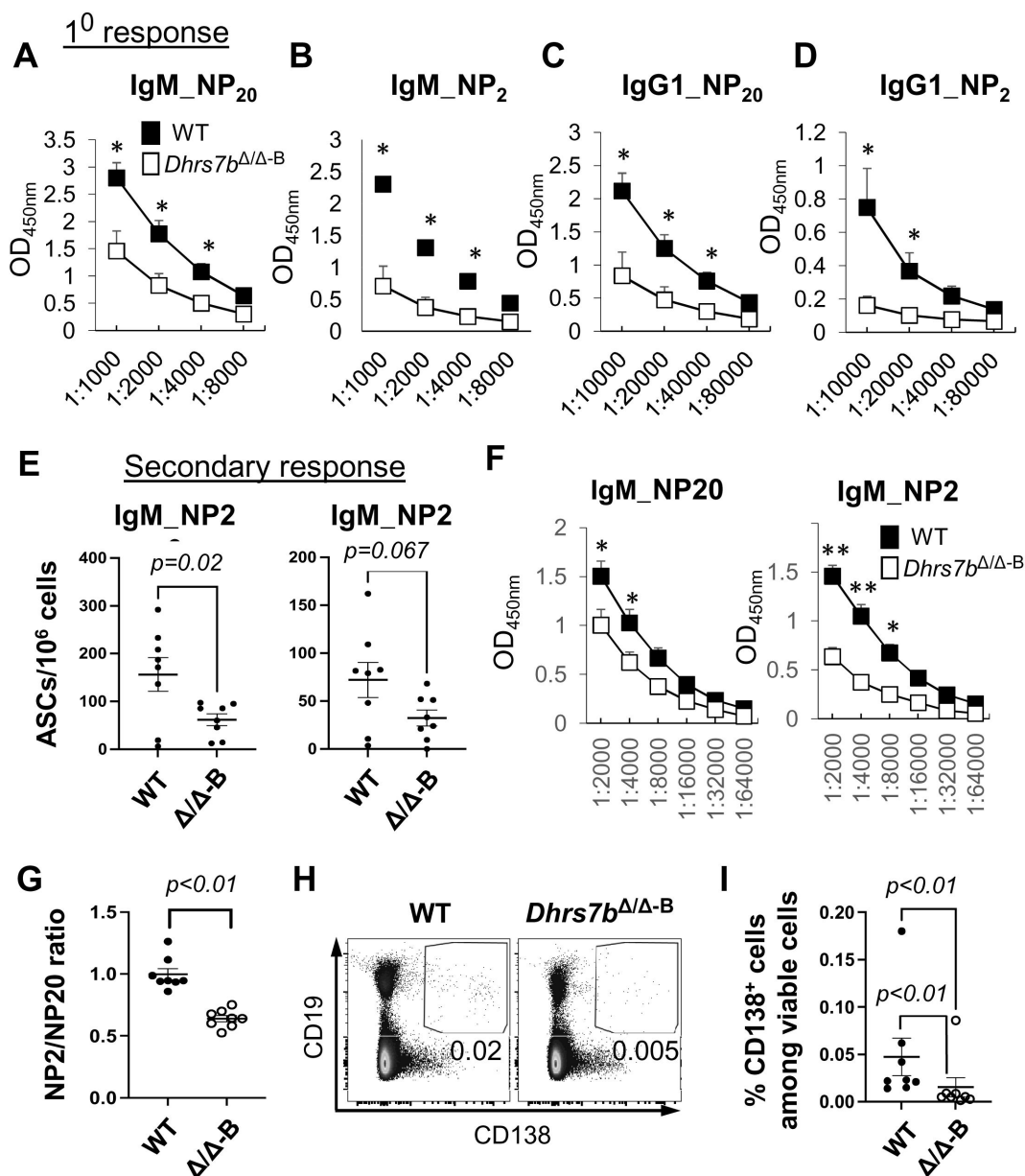


Figure 6.

Ab response and affinity increase promoted by PexRAP in B cells.

Tamoxifen-treated mice (*huCD20-CreER^{T2}; Dhrs7b^{+/+}* or *huCD20-CreER^{T2}; Dhrs7b^{ff}*, i.e., *Dhrs7b^{Δ/Δ-B}*) were immunized as in Fig. 5, with venous blood collected to measure Ag-specific Ab in the 1^0 response just prior to a second immunization, followed by harvesting a week thereafter (2^0 response). (A–D) Ag-specific Ab in primary response sera, prior to the boost. Shown are the all- (NP₂₀) and high- (NP₂) affinity anti-NP IgM (A, B) and IgG1 (C, D), as indicated, with NP₂₀ a high hapten density to detect both low- and high-affinity Ab and NP₂ a low hapten density selective for high-affinity Ab. (E, F) Levels of anti-NP IgM detected using NP₂₀ and NP₂ for ELISpot (E) and ELISA (F) as described in *Materials and Methods*. Graphs show mean ($\pm$ SEM) number of Ab-secreting cells in spleen (E) and (F) mean ($\pm$ SEM) OD_{450nm} in measurement of NP-specific IgM in serial dilutions of sera. P values were calculated by students' t-test. * indicates $p<0.05$, and ** indicates $p<0.01$. (G) PexRAP in B cells promotes Ab affinity maturation. The bar graph shows the mean ($\pm$ SEM) ratios of high-affinity to all-affinity NP-specific IgM Ab in sera of individual mice (each dot representing one subject) using OD_{450nm} values at the 1:2000 dilution, with data from three independent experiments comprising 8 mice of each type (WT; *Dhrs7b^{Δ/Δ-B}*). P values were calculated by Mann-Whitney U test. (H, I) PexRAP promotes CD138⁺ cell differentiation in vivo. B cells were adoptively transferred into μ MT recipient mice and analyzed at 4 days after transfer. Representative flow plot of CD138 and cD19 expression (H) and aggregated frequencies of CD138⁺ CD19⁺ cells in the viable cell gate (I) from four independent replicate experiments. To test for a potential distortion arising from outlier values, statistical testing was performed both with and without their inclusion.

metabolites within GC B cells. To test if *Dhrs7b* functions within GC, we used conditional deletion of *Dhrs7b* driven by the *S1pr2-CreER^{T2}* transgene whose expression at high levels marks GC B cells (69). Of note, experiments with a fate-marking reporter allele showed that activated B cells that lack the GC B phenotype were not marked by this conditional Cre after immunization with the NP-carrier approach (69). *Dhrs7b^{ff}*; *S1pr2-CreER^{T2}* and control *S1pr2-CreER^{T2}* (control) mice were immunized with SRBC, and tamoxifen was injected at a time point after the initiation of GC to test more directly that the inactivation of *Dhrs7b* would be in GC B cells rather than pre-GC blasts (Figure 7A). Frequencies of GC B cells were substantially lower in tamoxifen-treated *Dhrs7b^{ff}*; *S1pr2-CreER^{T2}* mice (Figure 7B, C). While this finding does not exclude that there may be an additional effect of impaired clonal expansion in B cells activated by immunization but not yet resident in the GC, the results indicate that PexRAP functions within GC B cells to support a full population of this subset. All together, these data provide evidence that *Dhrs7b* expression influences the lipidome and function of GC B cells. Notably, an optimal GC response requires PexRAP function in B cells, and the B cell-intrinsic functions of *Dhrs7b* promote the quantity and affinity maturation of Ab responses, in part via net proliferation of B cells.

***Dhrs7b* contributes to modulation of ROS and their impact on B cell population growth**

Increased susceptibility to cell death after activation would be a mechanism that could cause the reduced population expansion, GC, and Ab production. Consistent with earlier analyses of LPS-stimulated B cells (70), we had noted that ROS levels were higher in GC B cells, memory B cells (MBCs) and ASCs than in the naive B2 B cell pool (36). ROS can be produced in activated B cells via diverse sources that include mitochondrial electron transport chain function, NADPH oxidase activated by BCR engagement, and fatty acid oxidation in peroxisomes (71–73). ROS can positively mediate signal transduction, but their steady-state concentrations need to be calibrated because overproduction can have deleterious effect on cells, for instance via lipid peroxidation at excessive rates (74–76). Endogenous antioxidant properties have been imputed to plasmalogens because the vinyl ether bond is susceptible to reaction with reactive oxygen, thereby scavenging ROS (48, 57–59), so we tested if *Dhrs7b* function contributes to redox control in B cells. ROS levels were ~2-fold higher in GC B cells from immunized *Dhrs7b^{Δ/Δ}* mice compared to those from immunized WT controls (Fig 7D). *Dhrs7b^{Δ/Δ}* GC B cells also showed higher cell death and attenuated BrdU incorporation (Fig 7E, F; Supplemental Fig 6C). Moreover, CD40-stimulated *Dhrs7b^{Δ/Δ}* B cells exhibited ~3 fold higher signal and about a doubling when stained with fluorescent sensors of cellular and mitochondrial ROS, respectively, compared to WT B cells *in vitro* (Fig. 8A–D). Substantially increased ROS also were measured after activation by BCR and CD40 cross-linking (Supplemental Fig 6D). Iron- and copper-dependent lipid peroxidation is considered to be a biological mechanism for ROS-mediated cell death (75, 76). Lipid peroxidation measured by a fluorescent indicator (C11-Bodipy) *in vitro* was higher in *Dhrs7b^{Δ/Δ}* B cells compared with WT controls after each type of mitogen activation (Fig. 8E; Supplemental Fig 6E). Chemical elicitation of increased ROS and lipid peroxidation in B cells, independent from B cell activation or proliferation, rapidly led to reduced cell viability and B cell numbers (Fig. 8F; Supplemental Fig 7A). Furthermore, *in vitro* analyses indicated that B cells lacking PexRAP also exhibited increases in the activated executioner caspase, cleaved caspase-3 (CC3) along with early apoptotic cells that are annexin V⁺ but exclude 7-aminoactinomycin D (Fig. 8G, H). These data indicate that *Dhrs7b* function supports not only protection against cell death, but also control of cellular and mitochondrial ROS levels and their impact on lipid modification (Supplemental Fig. 7). Of note, while H₂O₂ did not increase this early apoptotic population (Supplemental Fig. 7C), increasing ROS with menadione drove B cell death preceded by annexin V⁺ state without an increase in C11-Bodipy signal (Supplemental Fig. 7D–G). We infer that rather than a single mode of death, increased ROS resulting from PexRAP depletion probably stimulates at least two distinct modes of B cell death.

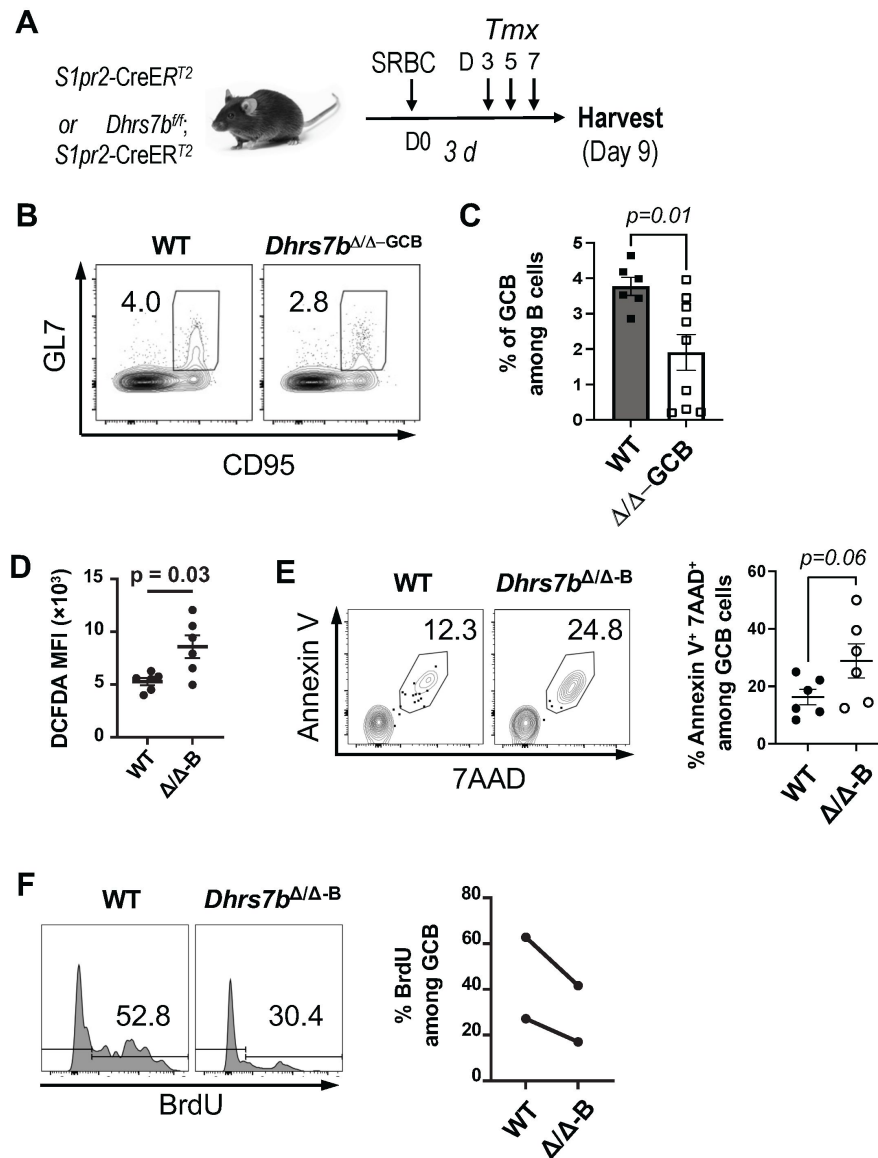


Figure 7.

Function of PexRAP in GC response.

(A) Schematic of the time line, with immunization followed later by tamoxifen injections into *S1pr2-CreER^{T2}* mice (*Dhrs7b^{+/+}* or *Dhrs7b^{fl/fl}*). Note that deletion is initiated only just as the germinal center reaction starts (~ 3.5 d post-immunization). Mice (*S1pr2-CreER^{T2}; Dhrs7b^{+/+}* or *S1pr2-CreER^{T2}; Dhrs7b^{fl/fl}*) were immunized with NP-OVA, treated with tamoxifen on days 3, 5, and 7 after NP-OVA immunization (36) and harvested at day 9. (B) Representative flow plots of GL7⁺CD95⁺ GC B cells among viable B cells and (C) aggregated mean (±SEM) frequencies of GC B cells from three replicate experiments (7 WT; 9 cKO mice). Mann-Whitney U test was used to calculate p values. (D-F) Effect of PexRAP on the levels of ROS (D), cell death (E), and proliferation (F) of GC B cells. (D) Total cellular ROS in IgD^{neg} CD38^{neg} GL7⁺ GC B cells were determined by flow cytometry after staining with surface markers and H₂DCFDA as described in the Methods. The graph shows the mean (±SEM) geometric MFI of H₂DCFDA from two independent replicate experiments (n = 6 WT and 6 cKO). (E) PexRAP promotes GC B cell survival. Shown are the representative flow plot (left), and a dot graph aggregating all experiments' outcomes for the frequencies of annexin V⁺ 7AAD⁺ cells in GC B gated cells as in Fig 7D (right panel). (F) PexRAP regulates proliferation of GC B cells. Tamoxifen-treated mice (*huCD20-CreER^{T2}; Dhrs7b^{+/+}* or *huCD20-CreER^{T2}; Dhrs7b^{fl/fl}*, i.e., *Dhrs7b^{Δ/Δ-B}*) were immunized with SRBC, and the mice were injected with BrdU as described in Methods. Shown are a representative histograms for WT and *Dhrs7b^{Δ/Δ-B}* GC B cells as indicated (left panel) and a graph indicating the aggregated result of each independent experiments (right panel) (x=2; n = 6 WT and 6 cKO).

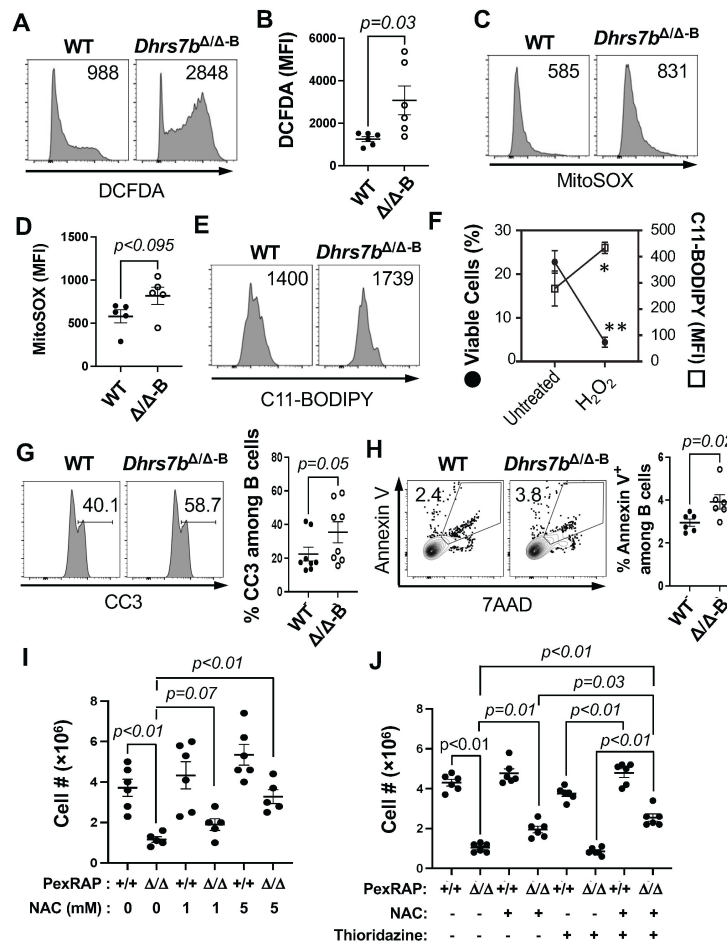


Figure 8.

PexRAP contributes to ROS homeostasis and B cell population growth in vitro.

(A-D) PexRAP is critical for maintenance of normal ROS levels. Bead-purified B cells from spleens of tamoxifen-treated *huCD20-CreER*^{T2} mice (*Dhhrs7b*^{Δ/Δ-B} and *Dhhrs7b*^{+/+}) were cultured 3 days in anti-CD40, BAFF, IL-4, IL-5 and 4-hydroxytamoxifen. Total cellular (A, B) and mitochondrial ROS, mtROS (C, D) in B lymphoblasts were then determined by flow cytometry after staining with surface markers and H₂DCFDA and MitoSOX, as described in the Methods. Representative histogram image of H₂DCFDA (A) and MitoSOX (C) in the B cell gate, and aggregated mean (± SEM) geometric MFI of H₂DCFDA (B) and MitoSOX (D) from 3 independent experiments, each using two mice of each type (6 WT; 6 cKO). P values were calculated by Mann-Whitney U test. (E) PexRAP restrains lipid peroxidation. B cells were activated and cultured as in (A). A representative result of flow cytometric analyses of lipid peroxidation assayed using C11-Bodipy is shown, based on three independent experiments. (F, G) PexRAP promotes B cell survival. (F) Shown are the mean (±SEM) frequencies of total viable 'events' (by FSC, SSC, and 7-AAD exclusion) in flow cytometry (filled circles) and MFI of Bodipy-C11 (open squares) after exposure to H₂O₂ (200 μM). (*, ** - p=0.03 and 0.003, respectively). (G-I) WT and *Dhhrs7b*^{Δ/Δ} B cells were cultured (2 d) in anti-CD40, BAFF, IL-4, IL-5 and 4-hydroxytamoxifen. (G) Increased cleaved caspase 3 (CC3) in PexRAP-deficient cells generated in vitro. Cleaved caspase 3 in B cells was detected by intracellular staining and flow cytometry. Shown are a representative pair of histograms for WT and *Dhhrs7b*^{Δ/Δ} B cells as indicated (left panel) and a dot graph aggregating all experiments' outcomes (right panel). (H) Frequencies of apoptotic B cells in cultures as in (A-D) were scored for annexin V and 7AAD as described (84). Shown are representative data of flow plots in the lymphocyte gate (left panel) and a dot graph aggregating all experiments' outcomes (right panel). (I, J) PexRAP and ROS control promote B cell population growth in vitro. WT and *Dhhrs7b*^{Δ/Δ} B cells were cultured 5 days in anti-CD40, BAFF, IL-4, IL-5 and 4-hydroxytamoxifen in the presence or absence of ROS scavenger NAC (1 mM vs 5 mM) (H). (I) B cells activated as in (A-G) were cultured 5 d in the presence or absence of NAC (1 mM) or thiordiazine (100 nM). The bar graphs show the mean (± SEM) recovered cell number from three independent experiments, each with two independent samples of each genotype (WT; *Dhhrs7b*^{Δ/Δ}). Complementary results of experiments including anti-IgM for BCR cross-linking are shown in **Supplemental Figs 2** and **6**.

Consistent with the increased steady-state ROS and death, activated *Dhrs7b*^{Δ/Δ} B cells exhibited a defect in B cell population growth *in vivo* and *in vitro* (Fig. 2; Fig. 8A-H). The well-established ROS scavenger, N-acetyl-L-cysteine (NAC), exerted a concentration-dependent effect that improved the survival and population growth of PexRAP-deficient B cells (Fig. 8I). We exploited the sub-maximal rescue by a lower concentration of NAC (1 mM) to explore if peroxisomal oxidative metabolism might contribute to the toxicity observed when B cells are PexRAP-depleted. Although thioridazine on its own provided no increase in the B cell population growth, its combination with 1 mM NAC treatment improved growth of the *Dhrs7b*^{Δ/Δ} B cell population compared to either agent on its own (Fig. 8J). Analyses of B cells activated by BCR crosslinking along with anti-CD40 also found that PexRAP expression in B cells supports their population increase (Supplemental Fig 6F). Moreover, inclusion of NAC in the cultures partially mitigated the inactivation of *Dhrs7b* (Supplemental Fig 6F). Of note, measurements of CTV partitioning identified a modest but statistically significant decrease in division counts of PexRAP-deficient B cells under these conditions, and found that NAC did not increase this component of proliferation (Supplemental Fig 6G, H). These data support the conclusion that sufficient generation of PexRAP-dependent products is crucial for B cell survival. Moreover, this enzyme contributes - directly or indirectly - to a resolution or detoxification of ROS that is critical for population growth of activated B cells.

***Dhrs7b* in B cells affects their oxidative metabolism and ER mass**

Our finding that mitochondrial ROS were increased prompted us to investigate how PexRAP affects physiological functions by measuring respiration in activated B cells. Mitochondrial stress tests with *in vitro* activated B lymphoblasts measured small but definite decreases in both basal and maximal respiration (oxygen consumption rates, or OCR) (Fig. 9A-C). The altered performance of mitochondria was associated with reductions in calculated ATP generation and proton leak as well as a small decrease in spare respiratory capacity (SRC) (Fig. 9D-F, respectively). In contrast, the rates of glucose-stimulated extracellular acidification - cytosolic reactions and a surrogate that can approximate glycolytic activity - were unaffected (Fig. 9G). Thus, the effects of PexRAP on B cell metabolism are not global, in that altered respiration did not reflect a decrease in glucose-stimulated acidification. Inasmuch as peroxisomes and mitochondria form contacts and functionally interact with the ER, we measured relative ER mass and found that the signal of the fluorophore ERTracker was consistently reduced in PexRAP-depleted B cells (Fig. 9H, I). We infer that both mitochondrial function and ER mass in B cells are promoted by the product of the *Dhrs7b* gene.

Discussion

We have shown herein that B lymphocyte expression of an enzyme essential for biosynthesis of a subset of ether phospholipids is crucial for part of the increases in their levels localized to GC. Moreover, the genetic approach used to deplete B cells of PexRAP allowed us to show that the B cell type-restricted gene, *Dhrs7b*, promotes B cell survival, proliferation, and GC size along with affinity maturation and serum concentrations of Ag-specific Ab after immunization. Taken together, this work (i) establishes the predictive value of the discovery approach that identified an unexpected feature of cellular biochemistry in GC, (ii) indicates that B cell-autonomous generation of ether lipid species is crucial for B cell survival, and (iii) regulates the qualities of the Ab elicited by immunization.

Using imaging mass spectrometry for discovery-based hypothesis generation, we had reported that the concentrations of at least a dozen ether phospholipids are increased in splenic GC relative to the remainder of the tissue (45). The analyses presented here confirm and extend the observations, and provide evidence that after immunization, these concentrations increase even in the primary B cell follicle, albeit to a modest degree when compared to the magnitude of increases in GC, and in B cells activated *ex vivo*. The cell type-specific gene inactivation after

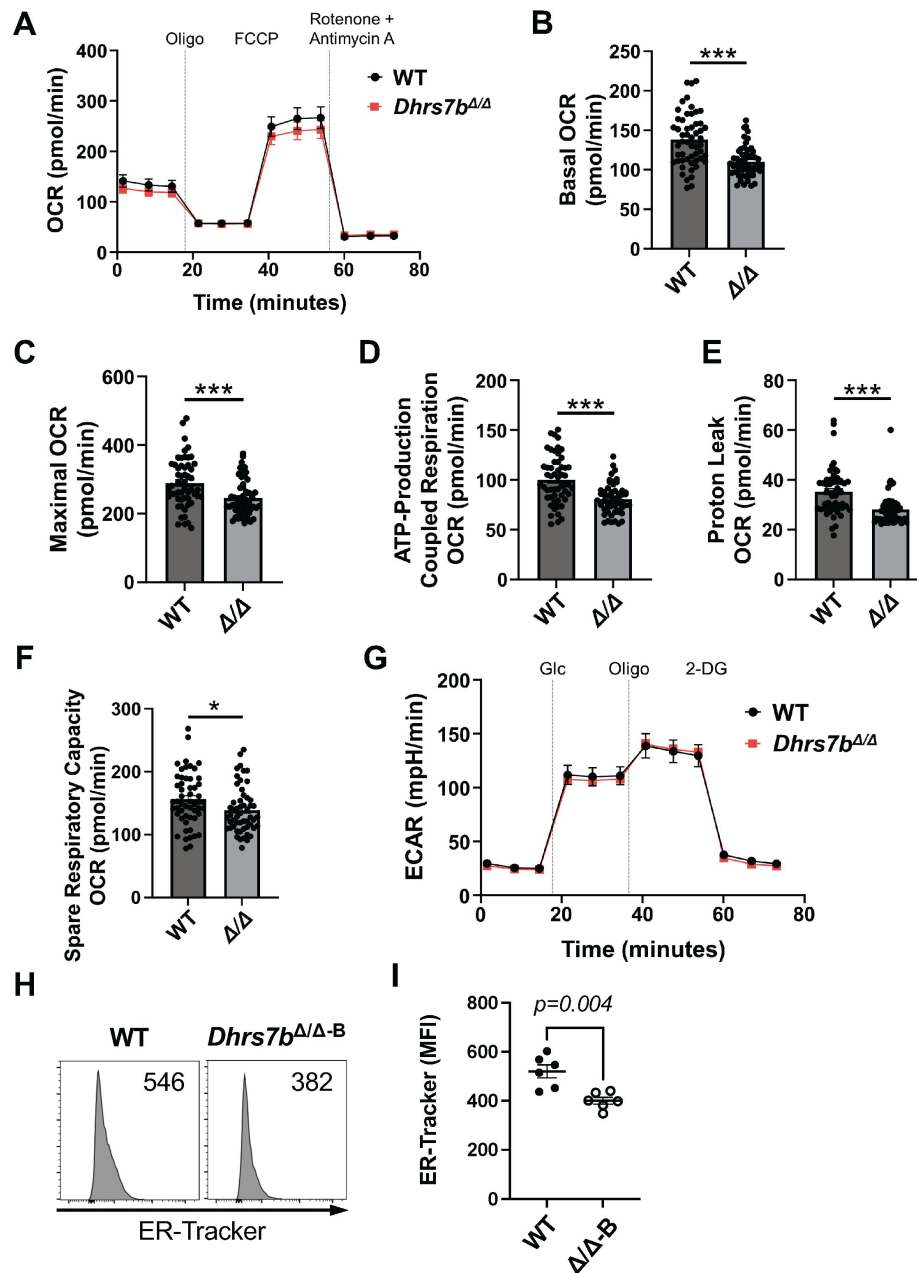


Figure 9.

PexRAP deficiency in activated B cells reduces mitochondrial metabolism and ER mass.

(A-G) Purified B cells were activated with and cultured (2 d) in anti-CD40, BAFF, IL-4, IL-5, and 4-OHT, then analyzed using a Seahorse XFe96 after harvest and division in equal portions. (A) Shown are aggregated results from three independent experiments measuring oxygen consumption rate (OCR) quantified via metabolic flux analysis during mitochondrial stress testing. (B) Basal OCR, (C) maximal OCR, (D) ATP-production coupled respiration, (E) proton leak, and (F) spare respiratory capacity of WT and *Dhrs7b*^{Δ/Δ} B cells assayed in (A). Data points of individual samples are shown (each individual dot), as well as bars to display mean (±SEM) values. T-tests with Welch's correction were used to calculate p-values. *, p<0.05; ***, p<0.001. (G) Extracellular acidification rate (ECAR) of B cells cultured as described previously (36) and assessed using the glycolytic stress test. (H, I) PexRAP influences ER mass. WT and *Dhrs7b*^{Δ/Δ} B cells were cultured 5 days in anti-CD40, BAFF, IL-4, IL-5 and 4-hydroxytamoxifen followed by the staining with ER-Tracker Green. Representative histogram of ER-Tracker Green in viable cells (H) and aggregated MFI (±SEM) of ER-Tracker Green from three replicate experiments (I). Mann-Whitney U test was used to calculate p values.

establishment of a pre-immune population establishes that an enzyme crucial for generation of a number of plasmalogens has substantial effects on B lymphocyte physiology and function. Plasmalogens and other ether lipids and phospholipids have long been shown to be major constituents of lipid bilayers and cell membranes (48, 49, 57). However, remarkably little is known about whether or not cell-intrinsic synthesis of ether lipids or the balance among their specific molecular species matters for hematopoietic cells or in immunity. Several inborn errors of metabolism in humans are attributable to loss-of-function mutations of genes encoding peroxisomal proteins that impact ether lipid (and hence plasmalogen) synthesis (58) but affect other critical processes as well [reviewed in (48, 58–62)]. Several of these - such as Zellweger Syndrome and rhizomelic chondrodysplasia punctata (RCDP) - and the mouse models generated to study mechanisms in these human disease, lead to severe neurological defects and early post-partum death (57, 58). Whether or not lymphocyte development, homeostasis or function were affected in these studies is not clear. Deficient generation of ether lipids is among many abnormalities caused by elimination of fatty acid synthase (FAS) (46, 47). Acute inactivation of *Fasn*, the gene encoding this enzyme, in young mature mice caused a decrease in spleen size disproportionate to the reduction in neutrophils, which was the most profound hematological consequence (46). In addition, circulating lymphocytes were reduced despite normal steady-state representation in the marrow. In-trans cell-extrinsic functions of ether lipids have been noted previously - specifically, as ligands for the invariant natural killer T cell receptor (77) or for a G-protein coupled receptor on natural killer cells (78). As such, the widespread loss of FAS function and the potential for indirect and pleiotropic effects of this enzyme deficiency left unresolved what cells actually depend on their own synthesis of ether lipids.

PexRAP, the enzyme encoded by the *Dhrs7b* gene, generates intermediates vital for the synthesis of some - but not all - ether lipids in cells such as neutrophils (46). We interpret our lipidomic data as indicating that this is the case for B lymphocytes. Specifically, although a number of plasmalogen or other ether lipid species in the naive, activated, or, in IMS, GC B cells were lower as a consequence of cell type-specific gene inactivation, other phospholipids in these classes were unaffected. Analyses of neutrophil numbers and survival provided evidence that this enzyme can be crucial for a normal membrane composition. Moreover, PexRAP promoted viability of this very short-lived cell type whose membranes have a high plasmalogen content (46). The bloodborne population of lymphocytes was reduced in the *Rosa26-CreER^{T2}, Dhrs7b^{fl/fl}* mice (46), but the specific cell types were not reported and cells in circulation may not reflect the spleen or other organs. Of note, the interpretation that the neutrophil phenotype is due to insufficiency of ether lipids or plasmalogens has been questioned based on alternative gene disruption results (54). Several models of unconditional gene inactivation or mutation that drastically reduced the phospholipid end-products were noted to have normal neutrophils [summarized in (54), albeit at older ages (4 - 7 mo) than those used in (46)]. However, these caveats are subject to points articulated in (53). There may be age-related changes in the processes analyzed herein using young (age ~ 2 mo) mice (53). Moreover, cellular and organismal selection for adaptation variants makes the results of acute loss-of-function inherently different from unconditional loss and heterozygote parentage (53). The potential for toxicity induced by 4OHT activation of *CreER^{T2}* was among concerns mooted about the earlier study of neutrophils (54). This concern may have been extrapolated from papers that noted impairment in earlier hematopoiesis when *Rosa26-CreER^{T2}* was combined with very intense regimens of tamoxifen administration to infant mice (65, 79). Such studies used individual doses over 4-fold higher than those used herein, and delivered by gavage five times on a daily basis instead of three. Nonetheless, the IgM⁺ B cell population sizes in spleen and marrow were normal even in the face of reduced erythropoiesis and thymopoiesis (65). Moreover, the analyses herein compared B lineage-specific deletion to *CreER^{T2}±, Dhrs7b^{+/+}* mice identically dosed in parallel with the test subjects. Thus, our work indicates that PexRAP is essential for normal physiology and function of mature B lymphocytes, in particular after their activation.

The findings are most consistent with a capacity for PexRAP function to contribute towards B lymphocyte survival and effective proliferation by restraining both executioner caspase activation and lipid peroxidation due to excessive ROS levels. For instance, population expansion of *Dhrs7b* Δ/Δ B cells was severely undermined after mitogenic stimulation in vitro with little difference in the division history but higher frequencies of annexin V-positive B cells. These findings are consistent with those obtained with neutrophils (46 [46](#)). On the surface such results might appear to present a conundrum relative to work indicating that both fatty acyl-CoA reductase 1 and the enzyme that later generates an alkenyl bond at a terminal step in plasmalogen biosynthesis are crucial for the susceptibility of HT-1080 and 786-O tumor cell lines to ferroptosis (55 [55](#)). However, our data are consistent with published evidence that PexRAP is needed for a narrower subset of ether lipids than the broad requirement for FAS or fatty acyl-CoA reductases (46 [46](#)–49 [49](#), 80 [80](#), 81 [81](#)). Moreover, molecular dissection of mechanisms in various non-hematopoietic cell types (including the cancer line 786-O) showed that ether lipid biosynthesis can either promote susceptibility or resistance to ferroptosis (82 [82](#)). Collectively, our findings suggest that for B cells the inactivation of PexRAP tilts the balance toward death and reduced Ab—perhaps involving lower fractions of poly-unsaturated fatty acids (PUFA) (82 [82](#)), or a lower ratio of plasmalogens (alkenyl linkages) relative to alkyl-linked ether lipids.

PexRAP localizes to peroxisomes (35 [35](#)–37 [37](#)), and prior work has suggested both that peroxisomes are increased in GC B cells and likely contribute to FAO by these cells (26 [26](#)). In line with the possibility that this organelle may affect PexRAP-dependent survival, our data indicated that specific inhibition of peroxisomal FAO could collaborate with sub-optimal ROS scavenging to mitigate the reduced growth of activated B cells. However, recent findings indicate that PexRAP can also be expressed in the ER and is capable of contributing to ether lipid biosynthesis via catalysis in this organelle in addition to or instead of the peroxisome (56 [56](#)). Thus, although ROS generated in peroxisomes likely contribute to the stress experienced by PexRAP-deficient B cells, the phenotypes observed herein may not exclusively reflect a need for the peroxisomal fraction of the enzyme but instead might include functions of ER-localized protein.

Our findings with an intermediary in ether lipid biosynthesis add to an increasing body of evidence that metabolism in B lymphocytes affects their differentiation or function (28 [28](#)–36 [36](#)). Many sources and forms of Ab can be protective, while in several autoimmune diseases these molecules can drive pathology (1 [1](#), 2 [2](#)). The capacity to generate increased affinities and circulating concentrations of Ab is an important factor both in protective immunity and autoimmune disease. Antigen-independent, TLR-driven PC differentiation requires increased oxidative phosphorylation (70 [70](#)), while oxidative metabolism in GC B cells promotes affinity maturation by unknown mechanisms (33 [33](#)–35 [35](#)). While not the exclusive determinant of such properties, GC are major sources of higher affinity Ab with a broader spectrum of specificities (9 [9](#), 13 [13](#), 14 [14](#)). Several lines of evidence suggest that PexRAP affects the physiology of GC B cells during their residence. Rates of cell cycling as measured with BrdU incorporation appeared reduced, while ROS levels within GC B cells were higher along with the frequency of death. Because GC Tfh cells provide new stimulation to GC B cells that present peptide recognized by the TCR, some of these effects may be secondary consequences of a substantially lower GC-Tfh population even when B cell-specific loss-of-function was tested. A speculative corollary is that the observed reduction in ER mass might be functionally notable. For instance, cognate B-T interaction in the GC involves BCR-mediated internalization of Ag followed by processing and MHC-II presentation of T cell epitope peptide. Moreover, ether lipids and plasmalogens are presented to innate-like T cells via CD1d (77 [77](#)). In addition to the GC-localized abnormalities, GC formation, size, and function also depend on the efficiency of population growth for B cells in a phase before they enter and take on characteristics of GC B cells, and then homeostatic and differentiative processes while in the secondary follicle. Thus, while the findings with the post-immunization inactivation of *Dhrs7b* using *S1pr2*-CreER^{T2} provide evidence that the expression of PexRAP is important within GC B cells, the decreases of GC and affinity-matured Ab are likely also to involve a pre-GC phase of

clonal expansion. In any case, the rapidity of the effects suggests that pharmacological inhibition of PexRAP may be a target in inflammatory disease in which B cells participate, particularly in light of the potential for impacts on pro-inflammatory neutrophils.

Materials & methods

Reagents

Monoclonal antibodies (mAb) against mouse CD40, CD90.2, B220 (CD45R), and other mAbs (purified, biotinylated, or fluorophore-conjugated) were from BD Biosciences or Tonbo Biosciences (San Diego CA) unless otherwise indicated. BAFF was from AdipoGen (San Diego, CA). Recombinant mouse IL-4 and recombinant mouse IL-5 were from Peprotech (Rocky Hill NJ). Glucose, 2-deoxyglucose, oligomycin, rotenone, antimycin A, carbonyl cyanide 4-phenylhydrazone (FCCP), N-acetyl-cysteine (NAC), thioridazine hydrochloride, H₂O₂, and 4-hydroxytamoxifen (4-OHT) were from Sigma-Aldrich Chemicals (St. Louis MO), and menadione from Cayman Chemicals (Ann Arbor, MI). Tamoxifen (Tmx) was from APExBio Technology (Houston TX). NP-BSA and NP-PSA (for capture in ELISA) as well as NP-OVA (for immunization) were obtained from Biosearch Technology (Novato CA). SRBCs (sheep red blood cells) were from Thermo Fisher Scientific (Waltham MA). CellTrace™ Violet cell proliferation kit, CM-H₂DCFDA, MitoSOX™ Red, and BODIPY™ 581/591 C11 were obtained from Invitrogen (Waltham, MA).

Mice, immunizations, ELISA and ELISpot

All animal protocols were reviewed and approved by the Vanderbilt University Institutional Animal Care and Use Committee. Mice were housed in ventilated micro-isolators under Specified Pathogen-Free conditions in a Vanderbilt University Medical Center mouse facility and used both male and female mice at 6–8 weeks of age. For cell type specific inactivation of *Dhrs7b* genes, *Dhrs7b^{fl/fl}* mice (46) were crossed with *Rosa26-CreER^{T2}* (46), *huCD20-CerER^{T2}* (30), or *S1pr2-CreER^{T2}* strains (51, 52), all on C57/BL6 genetic background. Tamoxifen was administered as previously reported (25, 30). To control for potential Cre toxicity, *CreER^{T2}* mice were similarly injected to use as wild-type (WT) controls. Mice (ages ~ 8wk) were immunized with SRBCs (2×10⁸ cells per mouse) and analyzed 1 week after immunization as described (25, 45). Alternatively, mice were immunized and boosted by i.p. injections [each of 100 µg NP₁₆-OVA (Biosearch Technologies, Novato, CA) emulsified in 100 µL of Imject™ Alum (Thermo Fisher Scientific, Pittsburgh, PA)] as described previously (25, 30), and harvested for analyses 1 wk after the 2nd injection. Isotype-specific relative levels of Ag-specific Ab were quantitated by capture ELISA using Ag (NP₂₀-BSA or NP₂-PSA for all- or high-affinity hapten-specific Abs, respectively) followed by SBA Clonotyping System (Southern Biotech, Birmingham AL), as described previously (25, 30). As previously described (30), frequencies of Ab-secreting cells (ASCs) were analyzed by ELISpot and quantitated using an ImmunoSpot Analyzer (Cellular Technology, Shaker Heights OH).

Cell cultures, proliferation assay, and reversion of population growth

Splenic B cells were purified (90-95%) using negative selection as previously described (30) or by positive selection with anti-mouse B220 microbeads (Miltenyi Biotec, Auburn, CA). B cells (5 × 10⁵ cells in 1 ml) were activated with anti-CD40 (1 µg/mL) and cultured with BAFF (10 ng/mL), IL-4 (10 ng/mL), IL-5 (10 ng/mL) and 4-OHT (50 nM) in Iscoves Modified Dulbecco's Medium (IMDM) supplemented with 10% Fetal Bovine Serum (FBS), 100 U/mL penicillin 100 µg/mL streptomycin (Invitrogen), 3 mM L-glutamine, nonessential amino acids (Invitrogen), 0.1 mM 2-ME (Sigma-Aldrich). To analyze the effect of ROS scavenger and inhibition of peroxisomal lipid oxidation on population growth, bead-purified B cells from tamoxifen-treated *huCD20-CerER^{T2}* or *Dhrs7b^{fl/fl}*; *huCD20-CerER^{T2}* mice were activated with anti-CD40, cultured with BAFF, IL-4 and IL-5 in the

presence or absence of NAC (1 mM and 5 mM) and/or thioridazine·HCl (100 μ M) for 5 days. For enhancement of ROS in WT B cells, purified B cells were analyzed 3 d after activation as for ROS scavenging, but vehicle, H₂O₂ (200 μ M) or menadione (8 μ M) was added at three different time points (overnight, 3 hours, and 1 hour prior to processing for flow cytometry to counting and measurements of DCFDA, annexin V, and C11 Bodipy signals).

For the analysis of proliferation in vitro, B cells were purified by depleting CD90.2⁺ T cells and CD138⁺ cells using biotinylated anti-CD90.2 Ab and biotinylated anti-CD138 Ab followed by streptavidin-conjugated microbeads (iMagTM; BD Bioscience), labeled with 5 μ M CellTraceTM Violet (CTV, Invitrogen), activated and cultured as above. To analyze proliferation in vivo, B cells were purified, labeled with CTV, and injected intravenously into B cell deficient μ MT recipient mice. Mice were harvested at 4 days after adoptive transfer. To measure proliferation rates of GC B cells, in vivo BrdU incorporation was performed by injecting mice with BrdU (2 mg/mouse; BD-Pharmingen, San Jose, CA) at 16 h and 4 h before harvesting spleens from SRBC-immunized mice. Cells were stained for surface markers (B220, IgD, GL7, CD38) followed by fixation, permeabilization, and staining with anti-BrdU-APC Ab using the BrdU Flow Kit (BD-Pharmingen) according to manufacturer's protocol.

Flow cytometry - general and measurements of ROS, lipids peroxidation, and ER

GC B cells were identified as GL7⁺ CD95⁺ events in the viable B220⁺ dump⁻ gate (dump channel consisting of one fluorophore for CD11b, CD11c, F4/80, Gr1, TCR β , and 7AAD), while plasmablasts were defined as B220⁺ CD138⁺ TACI⁺ dump⁻ cells. For flow analyses of total intracellular ROS and mitochondrial superoxide, B cells (1×10^6 cells) were washed with PBS and stained with 1.25 μ M CM-H₂DCFDA or 5 μ M MitoSOXTM Red in PBS (20 min at 37°C), respectively, then washed with 1% BSA containing PBS, and further stained with anti-B220, anti-CD19, anti-CD138, and Ghost-BV510. To measure the lipid peroxidation, cultured B cells (1×10^6) were washed with PBS and stained with 1.25 μ M C11-BODIPYTM in PBS (20 min at 37°C), and washed with 1% BSA containing PBS followed by surface staining as above. To analyze the level of endoplasmic reticulum (ER), B cells (1×10^6) were stained with 0.5 μ M ER-TrackerTM Green (Molecular Probes, Eugene OR) in Hank's Balanced Salt Solution with calcium and magnesium (HBSS/Ca/Mg; Gibco) for 30 min at 37°C.

For measurements of cell death (78 [DOI](#)), cultured B cells (1×10^6) were washed twice with PBS, once with Annexin V binding buffer (10 mM HEPES, pH 7.4, 140 mM NaCl, 2.5 mM CaCl₂), and then incubated (15 min at 20°C in the dark) with Annexin V-rPE, 7AAD, APC-conjugated CD138, APC-Cy7-conjugated B220, e450-CD19. Cells were then washed with Annexin V binding buffer and analyzed on the flow cytometer. To measure levels of cleaved caspase-3 (25 [DOI](#)), an activated apoptosis executor, the cultured cells were rinsed with PBS after harvest, stained (15 min at 4°C) with 7AAD, APC-conjugated CD138, APC-Cy7-conjugated B220, e450-CD19, washed (FBS, 1% v/v in PBS), then fixed with paraformaldehyde (4% w/v, 10 min at 20°C), permeabilized with permeabilization buffer (saponin, 0.2% w/v with FBS 1% in PBS), stained with FITC-conjugated Ab specific for cleaved caspase-3 (BD Biosciences), rinsed, and analyzed by flow cytometry.

Immunohistochemistry, MALDI-IMS, and IMS data analysis

Mice were immunized with SRBC and spleens were harvested at 1 week after immunization. Spleens were snap frozen on dry ice, and mounted to a cryostat chuck with a minimal amount of OCT (Thermo Fisher Scientific, San Jose, CA). Frozen spleens were cut at 12 μ m thickness using a Leica CM3050S (Leica, IL, USA) at -20°C, fixed with 1% paraformaldehyde for 10 min, washed twice with PBS, and blocked with M.O.MTM (Vector Lab) followed by incubation with GL7-FITC, anti-IgD-PE, and anti-CD35-biotin Ab followed by streptavidin-conjugated Alex647 at 4°C. After the tile scanning of spleen sections, GC size and numbers were quantified with FIJI Image J software.

For Matrix-Assisted Laser Desorption/Ionization (MALDI) Imaging Mass Spectrometry (IMS), frozen spleens were cut as above, and mounting onto indium tin oxide (ITO) coated microscope slides (Delta Technologies ETC). Slides were then vacuum desiccated for at least 30 minutes before matrix application. Pre-IMS autofluorescence (AF) images were acquired prior to matrix application on a Zeiss Axio Scan Z1 Slide Scanner using the brightfield channel and the DAPI (ex. 340-380 nm; em. 435-485 nm, blue), EGFP (ex. 450-490 nm; em. 500-550 nm, green), DsRed (ex. 538-562 nm; em. 570-640 nm, red) filter cubes. After taking AF images, a custom in-house developed sublimation device was used to apply the matrices 2,5-dihydroxyacetophenone (DHA) and 1,5-diaminonaphthalene (DAN) (Sigma Aldrich, St. Louis, MO, USA) to tissue sections for positive and negative ion mode analyses, respectively (78 [↗](#)). For immunized AID-GFP mice, the fluorescence emissions identified GC localization within the section used next for IMS. Alternatively, two serial sections of immunized mice spleens were used, one for 2D-IMS and the contiguous section for immunohistochemistry (IHC). Fluorescence data were registered with the 2D-IMS visualizations of phospholipids to align images (45 [↗](#)), allowing quantitation of the ion intensities in specific m/z peaks within defined micro-anatomic portions of the spleen (Fig. 4B [↗](#)). MALDI IMS data were acquired with a 10 μm pixel size (laser spot size 8 μm) in full scan mode using a Bruker trapped ion mobility time-of-flight (timsTOF) Flex mass spectrometer (Bruker Daltonics Billerica, MA, USA). Data were acquired with 250 shots per pixel and a mass range of m/z 200-2000. SCI₂S (software Bruker Daltonics) was used to process the data, normalize ion intensity, visualize ion images, and merge the images.

LC-MS for lipidomics

To prepare samples, a one-phase method was used to extract lipids (85 [↗](#)). Briefly, 0.5 mL of MeOH/MTBE/ CHCl_3 mix (1.3:1:1) was added to a frozen pellet of B-cells (1×10^6 cells total), spiked with 10 μL EquiSPLASH-lipidomics internal standard mix (Avanti Research), briefly vortexed and shaken gently for 20 min, followed by centrifugation at $20,000 \times g$ for 15 min at 10°C . The supernatant was transferred to a clean Eppendorf tube, evaporated under a gentle stream of N_2 gas, and resuspended in 100 μL methanol/ CHCl_3 (9:1) and 2 μL were used for LC-HRMS (high-resolution MS) analysis. Each sample was injected two times - one injection in positive ESI mode followed by one in negative mode. Pooled QCs were injected to assess the performance of the LC and MS instruments at the beginning, in the middle and at the end of each sequence. Discovery lipidomics data were acquired using a Vanquish UHPLC (ultrahigh performance liquid chromatography) system interfaced to a Q Exactive HF quadrupole/orbitrap mass spectrometer (Thermo Fisher Scientific).

Chromatographic separation was performed with a reverse-phase Acquity BEH C18 column (1.7 mm, $2.1 \times 150\text{mm}$, Waters, Milford, MA) at a flow rate of 250 $\mu\text{L}/\text{min}$. Mobile phases were made up of 10 mM ammonium formate and 0.1% formic acid in (A) $\text{H}_2\text{O}/\text{CH}_3\text{CN}$ (40:60) and in (B) $\text{CH}_3\text{CN}/\text{iPrOH}$ (10:90). Gradient conditions were as follows: 0–1 min, B = 20 %; 1–8 min, B = 20–100 %; 8–10 min, B = 100 %; 10–10.5 min, B = 100–20 %; 10.5–15 min, B = 20%. The total chromatographic run time was 15 min. Mass spectra were acquired over a precursor ion scan range of m/z 200 to 1,600 at a resolving power of 60,000 using the following HESI-II source parameters: spray voltage 4 kV (3 kV in negative mode); capillary temperature 250°C ; S-lens RF level 60 V; N_2 sheath gas 40; N_2 auxiliary gas 10; auxiliary gas temperature 350°C . MS/MS spectra were acquired for the top-seven most abundant precursor ions with an MS/MS AGC target of 1×10^5 , a maximum MS/MS injection time of 100 ms, and a normalized collision energy of 15, 30, 40. High resolution mass spectrometry data were processed with MS-DIAL version 4.70 in lipidomics mode (86 [↗](#)). MS1, and MS2 tolerances were set to 0.01 and 0.025 Da respectively. Minimum peak height was set to 30,000 to decrease the number of false positive hits. Peaks were aligned on a quality control (QC) reference file with RT tolerance of 0.1 min and mass tolerance of 0.015 Da. Default lipid library was used (Msp20210527163602_converted.lbm2), solvent type was set to HCOONH_4 to match the solvent used for separation, and the identification score cut off was set to 80%. All lipid classes were made available for the search. After lipid identification was completed, MS-DIAL results were exported

into Excel and cleaned using minimum RSD for QC samples set to 20% and minimum ratio of QC to Blank set to 10. For species identified in both PexRAP-depleted (*Dhrs7b* Δ/Δ) B cells and controls, mean levels (areas under peak curves) and their variance were analyzed in Excel.

Metabolic flux analyses

Oxygen Consumption Rate (OCR) were measured using Seahorse XF96 extracellular flux analyzer (Agilent Technology, Santa Clara, CA) as described previously (30). Briefly, in vitro activated B cells were washed twice, resuspended in XF Base Media (Agilent Technologies) supplemented with 2 mM L-glutamine, and equal numbers of B cells (2×10^5) were plated on extracellular flux assay plates (Agilent Technologies) coated with 2.5 $\mu\text{g/mL}$ CellTak (Corning) according to the manufacturer's protocol. Before extracellular flux analysis, B cells were rested (25 minutes at 37°C, atmospheric CO_2) in XF Base Media. OCR and ECAR were measured before and after the sequential addition of 1.5 μM oligomycin, 0.5 μM FCCP and 0.5 μM rotenone/antimycin A.

Supplemental information

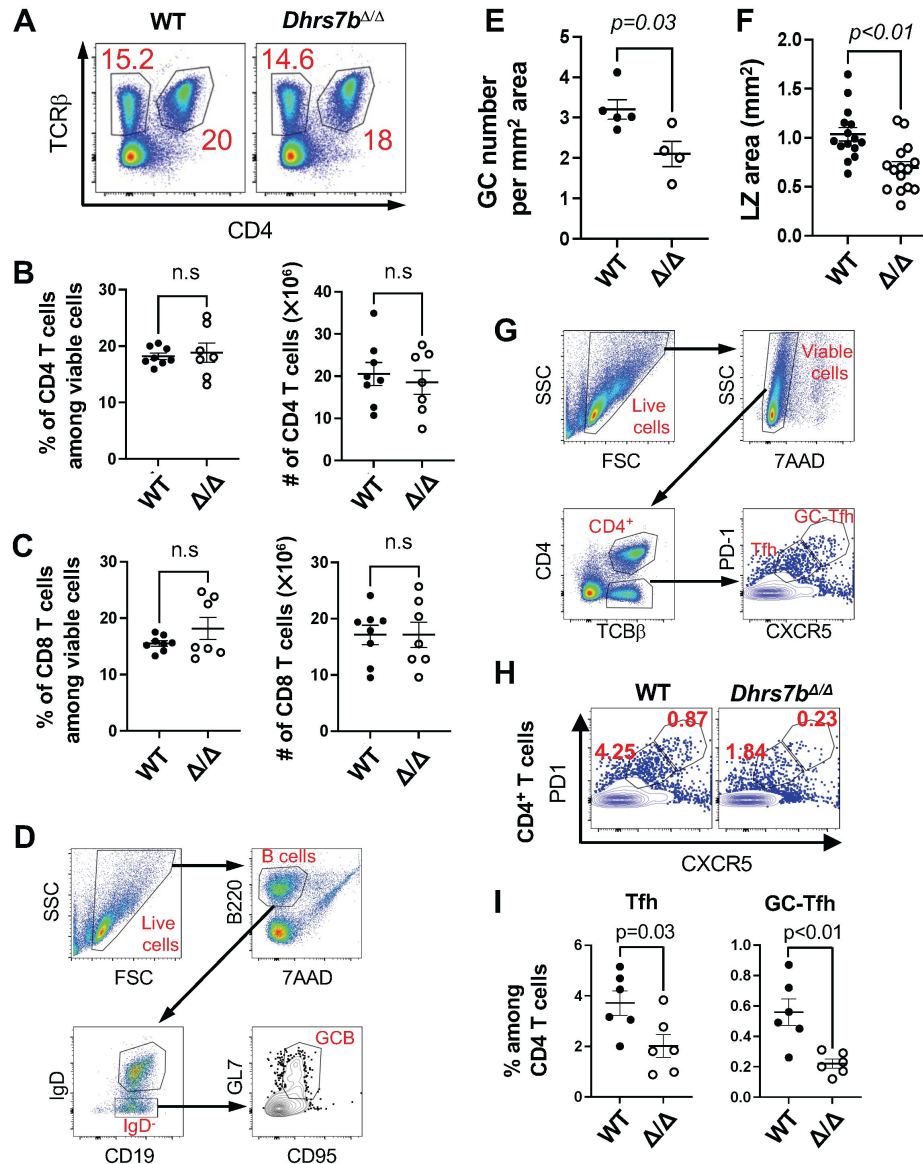
Supplemental Figures

Supplemental Table 1. Immunization-induced increases of selected ether lipids in GC. Shown are (a) a sample of m/z features characteristic of the indicated phospholipid species identified by the LIPIDMAPS database, and (b) the mean ($\pm$ SEM) ion intensity / counts in IMS data generated from spleens of AID-GFP mice, immunized or not (UI), after mapping to the indicated regions of interest (primary follicles or GC, i.e. secondary follicles) using fluorescent images. Shown are ions more accumulated in GC regions compared to primary follicles (8 ions from negative ion mode, and 5 ions from positive ion mode, all $p < 0.05$ for comparison of 1^0 follicle to UI (c), or GC to 1^0 follicle (d). P values were calculated by Mann-Whitney U test.

Supplemental Table 2. Impact of PexRAP on relative quantities of selected lipid species in primary and secondary follicles (GC). As in Table 1 except that B cells of immunized mice were either WT or lacked PexRAP (cKO, i.e., *Dhrs7b* Δ/Δ -B). Shown are (a) a sample of m/z features characteristic of the indicated phospholipid species. (b) the mean $\pm$ SEM ion intensity counts in IMS data after mapping to the indicated regions of interest as in Table 1. (c-f) indicate that $p < 0.05$ for the null hypothesis in considering a difference between WT mouse spleens after immunization versus UI controls (c), 1^0 B cell follicles in spleens of immunized WT vs *Dhrs7b* Δ/Δ -B mice (d), GC vs in 1^0 B cell follicles in spleens of immunized WT mice (e), and GC of immunized WT vs *Dhrs7b* Δ/Δ -B mice (f) in the ions counts for designated m/z features. P values were calculated by Mann-Whitney U test.

Data availability

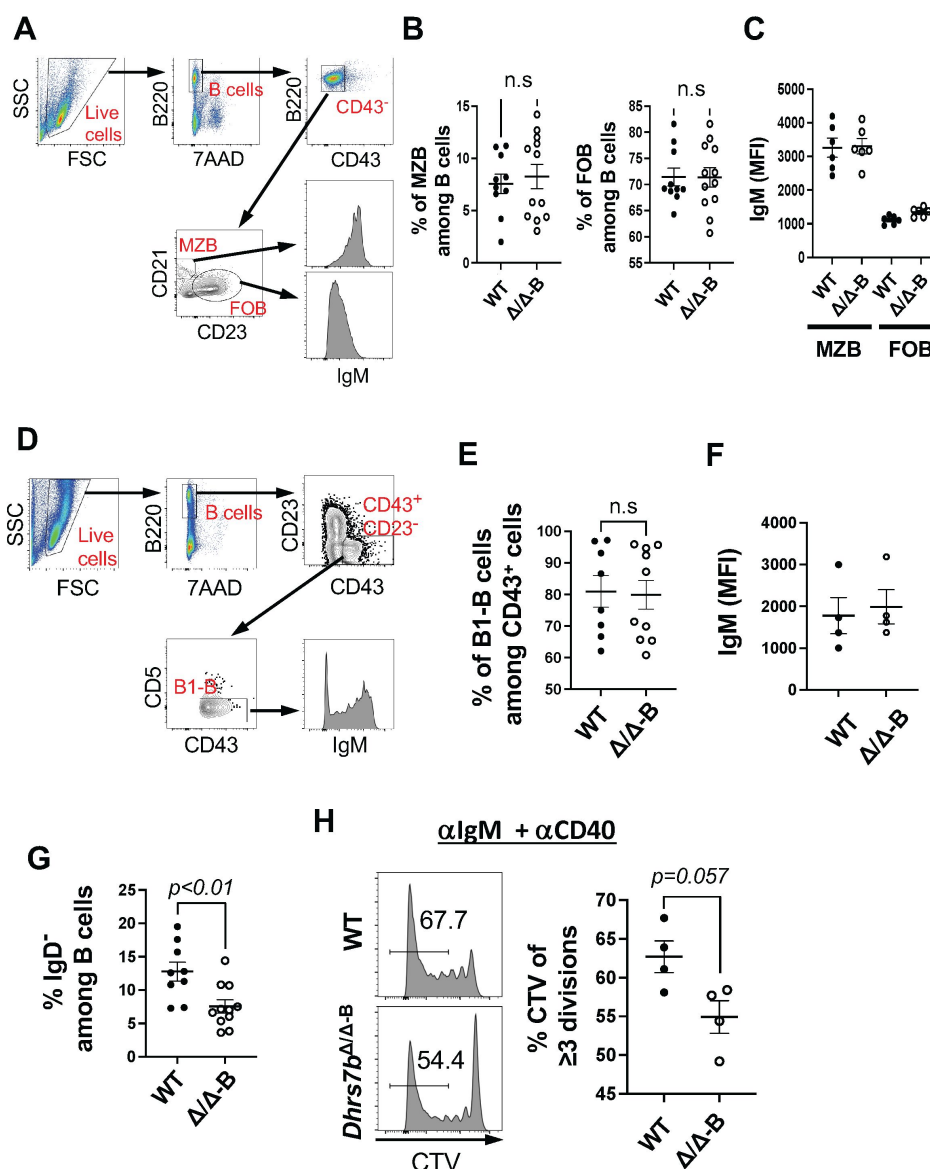
This work generated lipidomic and imaging mass spectrometric data for which repositories do not have a suitable format. Any / all such data are preserved by the Mass Spectrometry Research Center of Vanderbilt University using backed-up servers, and are freely available upon request. Other experimental data are backed up via Boothby lab devices and also available.



Supplemental Figure 1.

PexRAP is dispensable for steady-state T cell numbers, but impacts Tfh cell population.

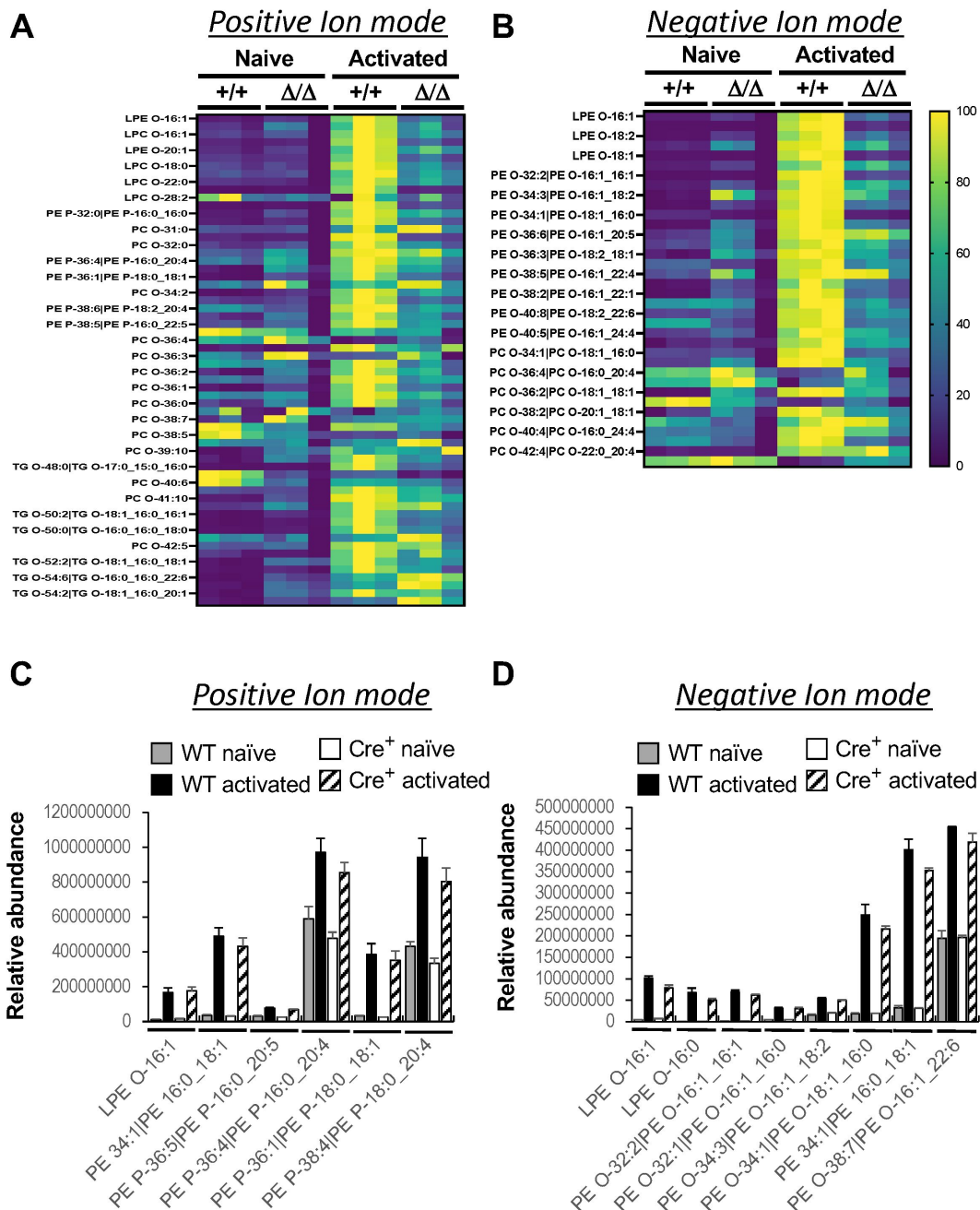
Results from flow cytometry analyses of splenocytes from mice (*Rosa26-CreER^{T2}*, *Dhrs7b^{+/-}*, "WT" or *Rosa26-CreER^{T2}*; *Dhrs7b^{fl/fl}*, "*Dhrs7b* Δ/Δ ") harvested after sequential tamoxifen treatments followed by immunization with SRBC and harvest 1 wk thereafter, all as described in *Materials and Methods* and illustrated in **Fig. 1A**. (A-C) Normal steady-state numbers of T cells. Shown are representative flow plots of splenic T cells (A), mean ($\pm$ SEM) frequencies of TCR β^+ CD4 $^+$ T cells (B), and CD8 (TCR β^+ CD4 $^-$) T cells (C) from WT and *Dhrs7b* Δ/Δ (cKO) mice based on three replicate experiments totaling 8 WT and 7 cKO mice. (D) Linking to **Fig. 1C-F**, representative gating scheme illustrating the flow-cytometric determination of splenic B cell frequencies and humbers (overall and GC B phenotype) in mice immunized as in A-C, **Fig. 1A**. (E, F) Linking to **Fig. 1G-I**, frequencies of GC, plotted as GC/mm 2 (E) and sizes of their LZ (plotted as area of each GC in the microscopic sections) (F) identified by immunofluorescent staining and microscopy of spleen sections from immunized mice. (G) Representative gating scheme illustrating the flow cytometric determination of Tfh and GC-Tfh cells in splenocyte suspensions of mice prepared and immunized as in **Fig. 1**. (H, I) PexRAP promotes Tfh cell population size. (H) Shown are representative flow plots of Tfh and GC-Tfh cells, determined using the gating scheme of **Supplemental Fig. 1G**. (I) Shown are mean ($\pm$ SEM) frequencies of PD-1 $^{\text{med}}$ CXCR5 $^{\text{med}}$ Tfh cells (left) and PD-1 $^{\text{hi}}$ CXCR5 $^{\text{hi}}$ GC-Tfh cells (right) from WT and cKO mice in two independent replicate experiments (n = 6 WT and 6 cKO).



Supplemental Figure 2.

PexRAP is dispensable for pre-immune B cell subset balance and surface IgM expression, but contribute to B cell population growth.

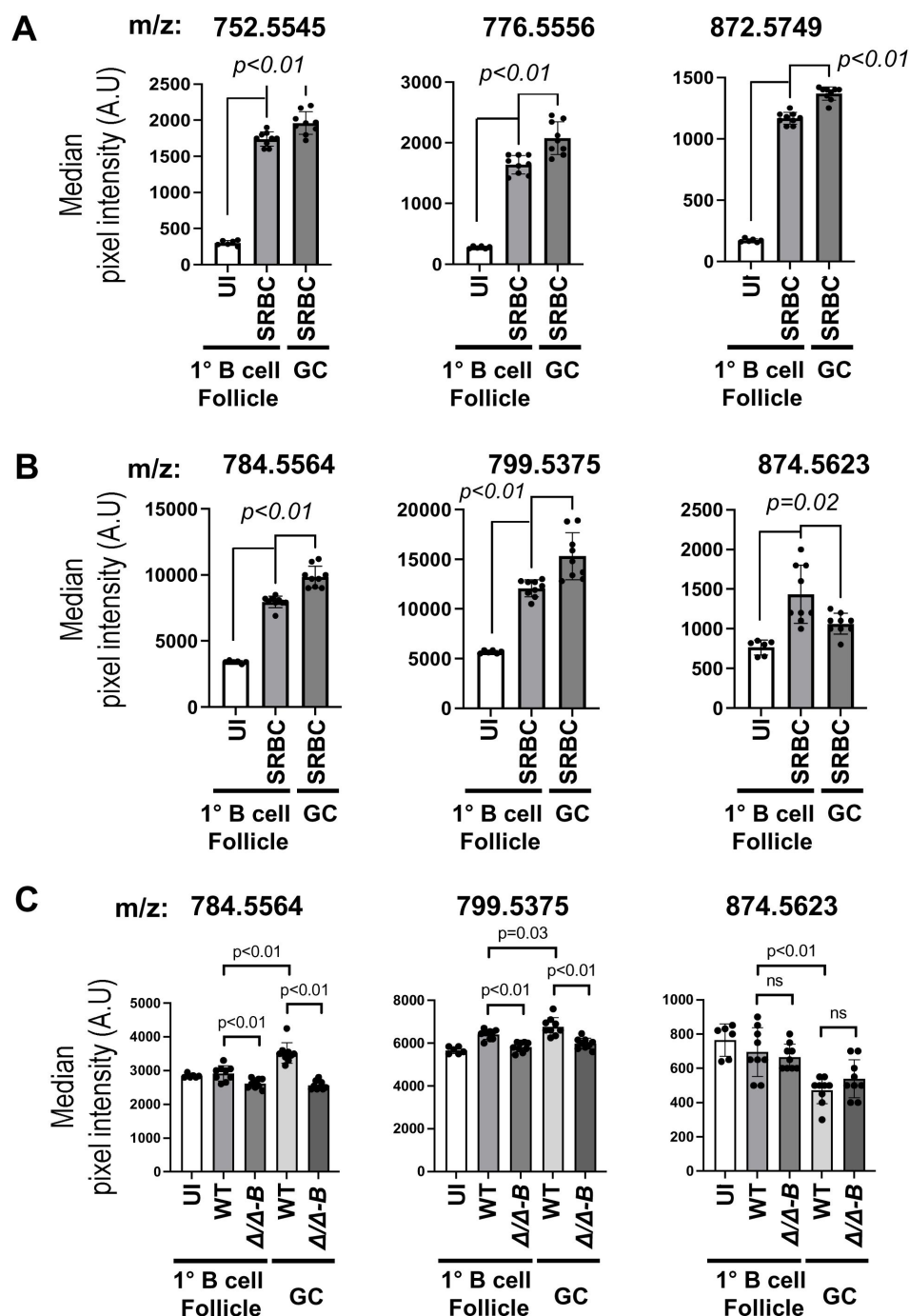
(A) A representative gating scheme illustrating the flow-cytometric estimation of the MZ and FO B cell populations in mice. (B, C) Shown are the mean ($\pm$ SEM) frequencies of B220⁺ CD43^{neg} CD23^{neg} CD21⁺ MZB cells (left panel) and B220⁺ CD43^{neg} CD23⁺ CD21^{lo} FOB cells (right panel) in spleen (B), and aggregated mean ($\pm$ SEM) geometric MFI of surface IgM on MZB and FOB subsets (C). (D) Gating strategy for identification of B1-b cells and measurement of their surface IgM levels. (E, F) Shown are the mean ($\pm$ SEM) frequencies of B220⁺ CD23^{lo} CD43⁺ CD5^{neg} B1-b cells in peritoneal cavity (E), and aggregated mean ($\pm$ SEM) geometric MFI of surface IgM on B1-b cells (F). (G) In vivo generation of IgD^{neg} progeny from transferred B cells of the indicated genotypes, as measured in four independent replicate experiments ($n = 8$ WT and 9 cKO). Shown are mean ($\pm$ SEM) frequencies of IgD^{neg} B cells as aggregate data for the frequencies of IgD^{neg} events in the gate for viable B cells recovered from μ MT recipient mice after transfer of WT or PexRAP-depleted B cells (experiments of Fig. 2D-F), as indicated, with representative flow plots in Fig. 2E. (H) PexRAP promotes B cell proliferation in vitro. B cells were purified and stained with CTV as in Fig. 2G. WT and $Dhrs7b^{\Delta/\Delta}$ B cells were stained with Cell Trace Violet (CTV), activated and cultured 4 days in anti-IgM, anti-CD40, BAFF, IL-4, IL-5, and 4-OHT. Shown are the representative flow-cytometric analysis of CTV partitioning (left), and aggregated frequencies of divided B cells from two independent replicate experiment, each using two separately sourced pools of B cells ($n = 4$ WT and 4 cKO). P values were calculated by Mann-Whitney U test.



Supplemental Figure 3.

Activation of B cells induces PexRAP-dependent increases in their ether and plasmalogen P-lipids.

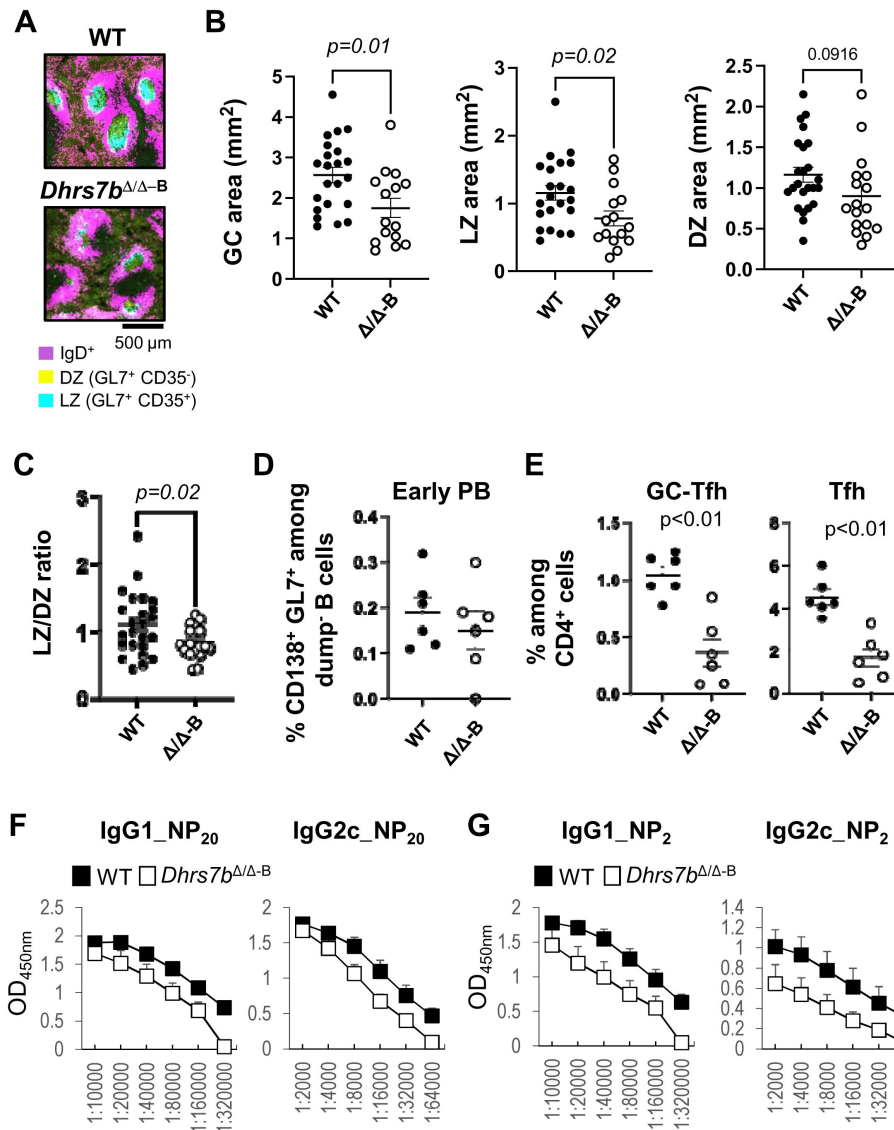
As in [Fig. 3](#), B lymphocytes of the indicated *Dhrs7b* genotypes (WT and Δ/Δ) were prepared from individual tamoxifen-injected mice and divided to analyze without activation ("naïve") or after activation and culture (48 h) ("activated"). Shown are heat maps with more extensive lists of ether phospholipids and plasmalogens identified in the LC-MS-MS analyses for both genotypes in positive (A) and negative (B) ion modes. (C, D) Controlling for transgene and tamoxifen effects. Shown are LC-MS-MS results for lipids in B cells, activated or not, as indicated, of conventional non-transgenic B6 mice not injected with tamoxifen and those of tamoxifen-injected CreER^{T2+} mice (the "+/+" samples of panels A, B and of [Figure 3](#)). Bar graphs in (C) and (D) show results of these control comparisons for some of the phospholipids analyzed in prior figures panels ([Fig. 3A, B](#)).



Supplemental Figure 4.

Immunization induces increases in P-lipids of both primary and secondary follicles dependent on PexRAP in B cells.

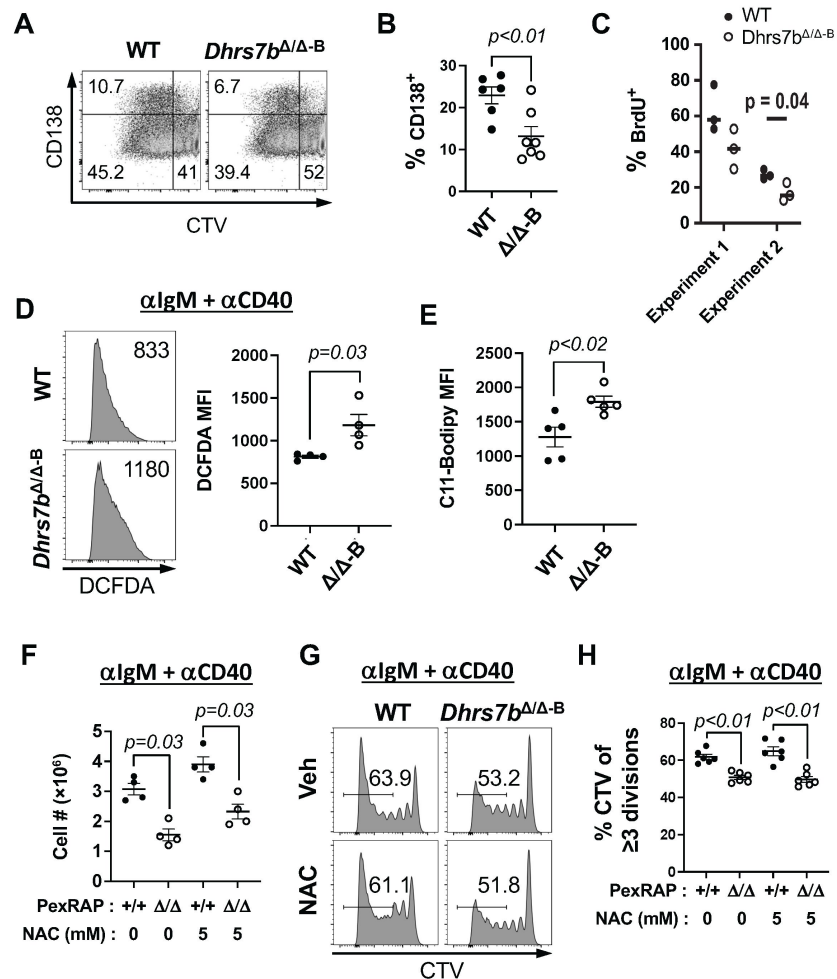
(A, B) Selected representative *m/z* features Spleens of AID-GFP were harvested at 7 days after immunization with SRBC and analyzed for exact mass signatures of ether lipid species in (A) negative and (B) positive ion modes. (C) Spleens of immunized mice (tamoxifen-treated *huCD20-CreER^{T2}*; *Dhrs7b^{f/f}*, i.e., *Dhrs7b^{Δ/Δ-B}*, and *huCD20-CreER^{T2}*, i.e., "WT", controls) were analyzed by IMS. Shown are the positive ions presented in (B) for GC vs follicles of immunized AID-GFP mice. The bar graph shows the mean (± SEM) ion intensities in lymphoid follicles and GC from WT and *Dhrs7b^{Δ/Δ-B}* spleens. Median intensity of each ion was obtained from three follicles per spleen and three GC per spleen from WT and *Dhrs7b^{Δ/Δ-B}* mice (three biological replication experiments totaling seven WT and six cKO subjects). P values were calculated by Mann-Whitney U test.



Supplemental Figure 5.

A subset of ether phospholipids in primary follicles and GC (secondary follicles) depend on PexRAP in B cells.

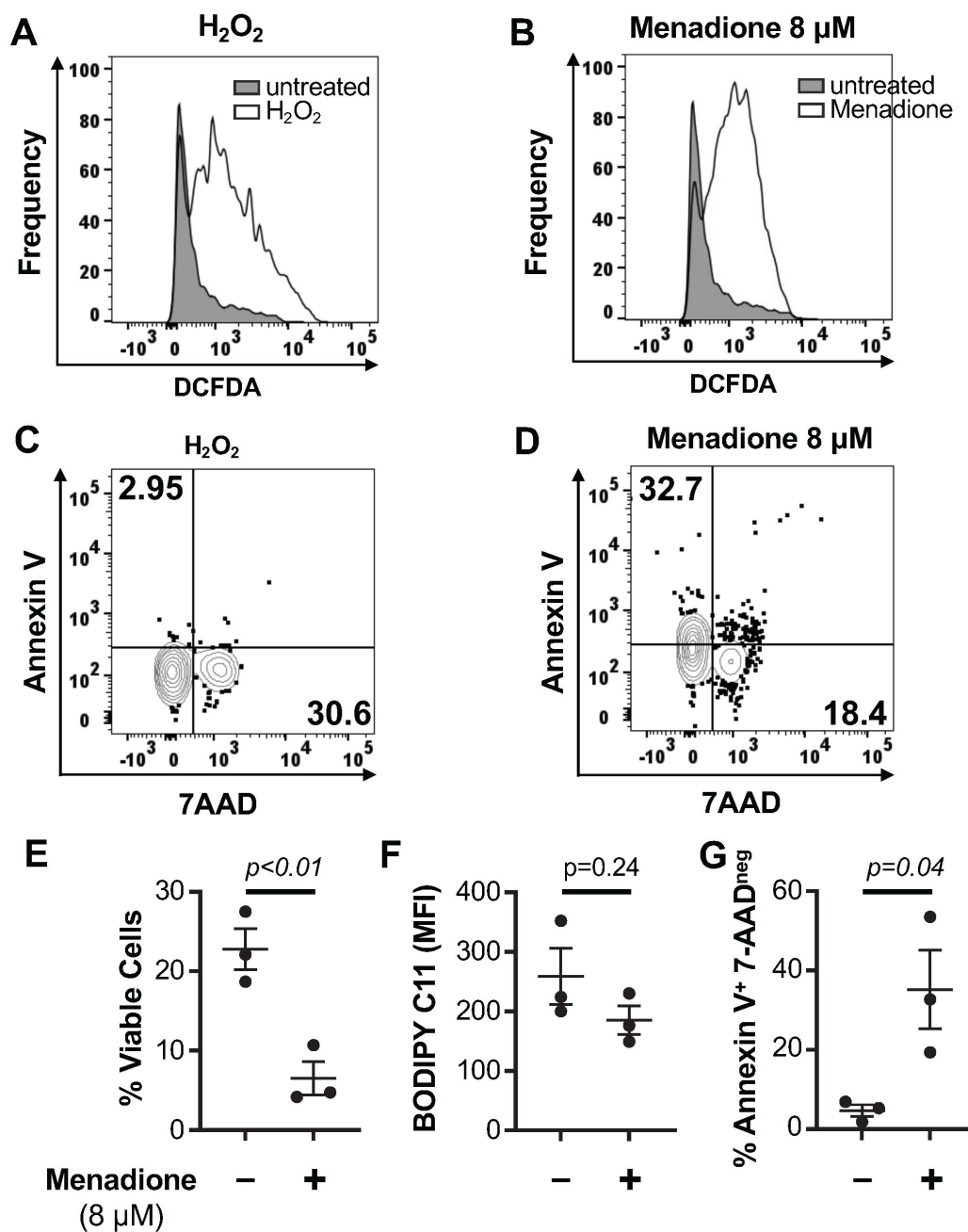
(A) As in **Fig. 4**, *Dhrs7b* was inactivated by tamoxifen injections into mice bearing the huCD20-CreER^{T2} transgene, followed by immunization with SRBC and analysis. Shown are representative higher-magnification images defining the primary (IgD⁺) and secondary (IgD^{lo/neg}, GL7⁺) follicles and the identification of light (CD35⁺) and dark (CD35^{neg}) zones. (B-C) After immunization and harvest of the mice as shown in **Fig. 4**, the sizes of GC, light zones (LZ), and dark zones (DZ) in photomicrographs of sectioned and immunofluorescently stained spleens of the immunized mice (using three GC per spleen, with seven WT and five *Dhrs7b*^{Δ/Δ-B}) were quantified. (B) Spleens of immunized mice (tamoxifen-treated huCD20-CreER^{T2}; *Dhrs7b*^{f/f} and huCD20-CreER^{T2} controls, as indicated) were analyzed by immunofluorescence staining as in (A). Each dot represents one follicle in a given mouse. (C) Dot graph showing the ratio of LZ/DZ using the data of panel B. (D, E) Mice (huCD20-CreER^{T2}; *Dhrs7b*^{f/f}, or huCD20-CreER^{T2}; *Dhrs7b*^{f/f}) were treated with tamoxifen and immunized with SRBC as in **Fig 4A**, and were analyzed 7 d after immunization. (D) Shown are aggregated frequencies of CD138⁺ GL7⁺ early plasmablasts in B220⁺ IgD^{neg} dump^{neg} gate (dump channel = IgD, CD11b, CD11c, F4/80, Gr1, TCRb, and 7AAD). (E) The aggregated frequencies of GC-Tfh (PD-1^{hi}, CXCR5^{hi}) and Tfh (PD-1^{int}, CXCR5^{int}) among CD4⁺ T cells with the gating strategy as shown in **Supplemental Fig. 1G**. (x=2; 6 WT vs 6 cKO). P value was calculated by Mann-Whitney U test. (F, G) Mean (±SEM) absorbances of ELISA performed across the indicated dilutions of sera from the tamoxifen-treated mice (huCD20-CreER^{T2}; *Dhrs7b*^{f/f} or huCD20-CreER^{T2}; *Dhrs7b*^{f/f}, i.e., *Dhrs7b*^{Δ/Δ-B}) of **Fig. 6**. Shown are the results for (F) all- (NP₂₀ captured) and (G) high- (NP₂ captured) -affinity anti-NP Ab of the IgG1 and IgG2c isotypes, as indicated.



Supplemental Figure 6.

PexRAP contributes to GC B cell proliferation, ROS homeostasis and B cell population growth.

(A, B) Proliferation and development of CD138⁺ progeny derived from B cells activated and cultured with anti-CD40, BAFF, IL-4, and IL-5. Shown are representative flow plots (A), and aggregated mean ($\pm$ SEM) frequencies of CD138⁺ cells in viable lymphocyte from three independent experiments with 6 WT and 7 *Dhhrs7b*^{Δ/Δ-B} mice at $t = 5$ dq. (C) PexRAP regulates proliferation of GC B cells. Tamoxifen-treated mice (*huCD20-CreERT2*; *Dhhrs7b*^{+/+} or *huCD20-CreERT2*; *Dhhrs7b*^{ff}, i.e., *Dhhrs7b*^{Δ/Δ-B}) were immunized with SRBC, and the mice were injected with BrdU as described in Methods. Shown are a dot graph aggregating all experiments' outcomes for the frequencies of BrdU⁺ cells in GCB cells ($x=2$; $n = 6$ WT and 6 cKO). (D) WT and *Dhhrs7b*^{Δ/Δ} B cells were activated and cultured 3 days in anti-IgM, anti-CD40, BAFF, IL-4, IL-5, and 4-OHT. Shown are representative histogram image of H₂DCFDA in the B cell gate (left panel), and aggregated mean ($\pm$ SEM) geometric MFI of H₂DCFDA (right panel) from two independent replicate experiment ($n = 4$ WT and 4 cKO). P values were calculated by Mann-Whitney U test. (E) PexRAP restrains lipid peroxidation. B cells were activated and cultured as in **Fig 8E**. Shown are the aggregated means ($\pm$ SEM) of geometric MFI of C11-Bodipy from two independent replicate experiments ($n = 5$ WT and 5 cKO). (F-H) NAC mitigated the impairment of population increase for PexRAP-deficient B cells, but failed to enhance cell division. B cells of the indicated genotypes were activated with combined BCR and CD40 cross-linking as in (D), cultured 5 d in the presence or absence of NAC (5 mM), counted and analyzed by flow cytometry. (F) The dot graph shows the mean ($\pm$ SEM) recovered cell number from two independent experiments. P values were calculated by Mann-Whitney U test. (G, H) Prior to activation, B cells of the each genotype were stained with CellTrace Violet (CTV). (G) Shown are flow cytometry outputs of CTV fluorescence in the viable B cell gate for the indicated samples from one of the independent replicate pools. A measurement bar denotes the cut-off between 0-2 divisions and ≥ 3 divisions, with the inset numbers representing the percentage of cells that divided at least three times. (H) A dot graph displaying the individual as well as mean percentages of B cells that divided at least three times for samples of the indicated genotypes and treatment conditions. P values were calculated by Mann-Whitney U test.



Supplemental Figure 7.

Distinct outcomes of heightened ROS elicited by H_2O_2 versus menadione.

(A-B) Induction of ROS independent from B cell activation. Shown are representative flow plots of DCFDA in the viable B cell gate after exposure (1 h) (A) to H_2O_2 (200 μM) or (B) menadione (8 μM), as in (87). (C-D) Representative flow plots of annexin V vs 7-AAD for identification of early-apoptotic B cells. Cultured B cells were analyzed 3 hr after treatment with (C) 200 μM H_2O_2 or (D) 8 μM menadione. Inset numbers represent the fraction of B cells that are Annexin V⁺ 7-AAD^{neg} or Annexin V^{neg} 7-AAD⁺. (E-G) Shown are the results of three independent experiments to measure (E) total cell viability, (F) BODIPY C11 MFI and (G) frequencies of Annexin V⁺ 7-AAD^{neg} frequency in B cells treated (8 μM) menadione overnight (E) or for 3 h (D). P-values were calculated using the unpaired Student's t-test.

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Reviewer #1 (Public review):

In this manuscript, Hoon Cho et al. present a novel investigation into the role of PexRAP, an intermediary in ether lipid biosynthesis, in B cell function, particularly during the Germinal

Center (GC) reaction. The authors profile lipid composition in activated B cells both in vitro and in vivo, revealing the significance of PexRAP. Using a combination of animal models and imaging mass spectrometry, they demonstrate that PexRAP is specifically required in B cells. They further establish that its activity is critical upon antigen encounter, shaping B cell survival during the GC reaction.

Mechanistically, they show that ether lipid synthesis is necessary to modulate reactive oxygen species (ROS) levels and prevent membrane peroxidation.

Highlights of the Manuscript:

The authors perform exhaustive imaging mass spectrometry (IMS) analyses of B cells, including GC B cells, to explore ether lipid metabolism during the humoral response. This approach is particularly noteworthy given the challenge of limited cell availability in GC reactions, which often hampers metabolomic studies. IMS proves to be a valuable tool in overcoming this limitation, allowing detailed exploration of GC metabolism.

The data presented is highly relevant, especially in light of recent studies suggesting a pivotal role for lipid metabolism in GC B cells. While these studies primarily focus on mitochondrial function, this manuscript uniquely investigates peroxisomes, which are linked to mitochondria and contribute to fatty acid oxidation (FAO). By extending the study of lipid metabolism beyond mitochondria to include peroxisomes, the authors add a critical dimension to our understanding of B cell biology.

Additionally, the metabolic plasticity of B cells poses challenges for studying metabolism, as genetic deletions from the beginning of B cell development often result in compensatory adaptations. To address this, the authors employ an acute loss-of-function approach using two conditional, cell-type-specific gene inactivation mouse models: one targeting B cells after the establishment of a pre-immune B cell population (Dhrs7b^{f/f}; huCD20-CreERT2) and the other during the GC reaction (Dhrs7b^{f/f}; S1pr2-CreERT2). This strategy is elegant and well-suited to studying the role of metabolism in B cell activation.

Overall, this manuscript is a significant contribution to the field, providing robust evidence for the fundamental role of lipid metabolism during the GC reaction and unveiling a novel function for peroxisomes in B cells.

Comments on revisions:

There are still some discrepancies in gating strategies. In Fig. 7B legend (lines 1082-1083), they show representative flow plots of GL7+ CD95+ GC B cells among viable B cells, so it is not clear if they are IgDneg, as the rest of the GC B cells aforementioned in the text.

Western blot confirmation: We understand the limitations the authors enumerate. Perhaps an RT-qPCR analysis of the Dhrs7b gene in sorted GC B cells from the S1PR2-CreERT2 model could be feasible, as it requires a smaller number of cells. In any case, we agree with the authors that the results obtained using the huCD20-CreERT2 model are consistent with those from the S1PR2-CreERT2 model, which adds credibility to the findings and supports the conclusion that GC B cells in the S1PR2-CreERT2 model are indeed deficient in PexRAP.

Lines 222-226: We believe the correct figure is 4B, whereas the text refers to 4C.

Supplementary Figure 1 (line 1147): The figure title suggests that the data on T-cell numbers are from mice in a steady state. However, the legend indicates that the mice were immunized, which means the data are not from steady-state conditions.

<https://doi.org/10.7554/eLife.104580.2.sa2>

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Summary:

In this study, Cho et al. investigate the role of ether lipid biosynthesis in B cell biology, particularly focusing on GC B cell, by inducible deletion of PexRAP, an enzyme responsible for the synthesis of ether lipids.

Strengths:

Overall, the data are well-presented, the paper is well-written and provides valuable mechanistic insights into the importance of PexRAP enzyme in GC B cell proliferation.

Weaknesses:

More detailed mechanisms of the impaired GC B cell proliferation by PexRAP deficiency remain to be further investigated. In minor part, there are issues for the interpretation of the data which might cause confusions by readers.

Comments on revisions:

The authors improved the manuscript appropriately according to my comments.

<https://doi.org/10.7554/eLife.104580.2.sa1>

Author response:

The following is the authors' response to the current reviews.

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In this manuscript, Hoon Cho et al. present a novel investigation into the role of PexRAP, an intermediary in ether lipid biosynthesis, in B cell function, particularly during the Germinal Center (GC) reaction. The authors profile lipid composition in activated B cells both in vitro and in vivo, revealing the significance of PexRAP. Using a combination of animal models and imaging mass spectrometry, they demonstrate that PexRAP is specifically required in B cells. They further establish that its activity is critical upon antigen encounter, shaping B cell survival during the GC reaction. Mechanistically, they show that ether lipid synthesis is necessary to modulate reactive oxygen species (ROS) levels and prevent membrane peroxidation.

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Overall, this manuscript is a significant contribution to the field, providing robust evidence for the fundamental role of lipid metabolism during the GC reaction and unveiling a novel function for peroxisomes in B cells.

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There are still some discrepancies in gating strategies. In Fig. 7B legend (lines 1082-1083), they show representative flow plots of GL7+ CD95+ GC B cells among viable B cells, so it is not clear if they are IgD^{neg}, as the rest of the GC B cells aforementioned in the text.

We apologize for missing this item in need of correction in the revision and sincerely thank the reviewer for the stamina and care in picking this up. The data shown in Fig. 7B represented cells (events) in the IgD^{neg} Dump^{neg} viable lymphoid gate. We will correct this omission/blemish in the final revision that becomes the version of record.

Western blot confirmation: We understand the limitations the authors enumerate. Perhaps an RT-qPCR analysis of the Dhrs7b gene in sorted GC B cells from the S1PR2-CreERT2 model could be feasible, as it requires a smaller number of cells. In any case, we agree with the authors that the results obtained using the huCD20-CreERT2 model are consistent with those from the S1PR2-CreERT2 model, which adds credibility to the findings and supports the conclusion that GC B cells in the S1PR2-CreERT2 model are indeed deficient in PexRAP.

We will make efforts to go back through the manuscript and highlight this limitation to readers, i.e., that we were unable to get genetic evidence to assess what degree of "counter-selection" applied to GC B cells in our experiments.

We agree with the referee that optimally to support the Imaging Mass Spectrometry (IMS) data showing perturbations of various ether lipids within GC after depletion of PexRAP, it would have been best if we could have had a qRT2-PCR that allowed quantitation of the Dhrs7b-encoded mRNA in flow-purified GC B cells, or the extent to which the genomic DNA of these cells was in deleted rather than 'floxed' configuration.

While the short half-life of ether lipid species leads us to infer that the enzymatic function remains reduced/absent, it definitely is unsatisfying that the money for experiments ran out in June and the lab members had to move to new jobs.

Lines 222-226: We believe the correct figure is 4B, whereas the text refers to 4C.

As for the 1st item, we apologize and will correct this error.

Supplementary Figure 1 (line 1147): The figure title suggests that the data on T-cell numbers are from mice in a steady state. However, the legend indicates that the mice were immunized, which means the data are not from steady-state conditions.

We will change the wording both on line 1147 and 1152.

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In this study, Cho et al. investigate the role of ether lipid biosynthesis in B cell biology, particularly focusing on GC B cell, by inducible deletion of PexRAP, an enzyme responsible for the synthesis of ether lipids.

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More detailed mechanisms of the impaired GC B cell proliferation by PexRAP deficiency remain to be further investigated. In minor part, there are issues for the interpretation of the data which might cause confusions by readers.

Comments on revisions:

The authors improved the manuscript appropriately according to my comments.

To re-summarize, we very much appreciate the diligence of the referees and Editors in re-reviewing this work at each cycle and helping via constructive peer review, along with their favorable comments and overall assessments. The final points will be addressed with minor edits since there no longer is any money for further work and the lab people have moved on.

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

In this manuscript, Sung Hoon Cho et al. presents a novel investigation into the role of PexRAP, an intermediary in ether lipid biosynthesis, in B cell function, particularly during the Germinal Center (GC) reaction. The authors profile lipid composition in activated B cells both in vitro and in vivo, revealing the significance of PexRAP. Using a combination of animal models and imaging mass spectrometry, they demonstrate that PexRAP is specifically required in B cells. They further establish that its activity is critical upon antigen encounter, shaping B cell survival during the GC reaction.

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study of lipid metabolism beyond mitochondria to include peroxisomes, the authors add a critical dimension to our understanding of B cell biology.

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Overall, this manuscript is a significant contribution to the field, providing robust evidence for the fundamental role of lipid metabolism during the GC reaction and unveiling a novel function for peroxisomes in B cells.

We appreciate these positive reactions and response, and agree with the overview and summary of the paper's approaches and strengths.

However, several major points need to be addressed:

Major Comments:

Figures 1 and 2

The authors conclude, based on the results from these two figures, that PexRAP promotes the homeostatic maintenance and proliferation of B cells. In this section, the authors first use a tamoxifen-inducible full Dhrs7b knockout (KO) and afterwards Dhrs7b^Δ/Δ-B model to specifically characterize the role of this molecule in B cells. They characterize the B and T cell compartments using flow cytometry (FACS) and examine the establishment of the GC reaction using FACS and immunofluorescence. They conclude that B cell numbers are reduced, and the GC reaction is defective upon stimulation, showing a reduction in the total percentage of GC cells, particularly in the light zone (LZ).

The analysis of the steady-state B cell compartment should also be improved. This includes a more detailed characterization of MZ and B1 populations, given the role of lipid metabolism and lipid peroxidation in these subtypes.

Suggestions for Improvement:

B Cell compartment characterization: A deeper characterization of the B cell compartment in non-immunized mice is needed, including analysis of Marginal Zone (MZ) maturation and a more detailed examination of the B1 compartment. This is especially important given the role of specific lipid metabolism in these cell types. The phenotyping of the B cell compartment should also include an analysis of immunoglobulin levels on the membrane, considering the impact of lipids on membrane composition.

Although the manuscript is focused on post-ontogenic B cell regulation in Ab responses, we believe we will be able to polish a revised manuscript through addition of results of analyses suggested by this point in the review: measurement of surface IgM on and phenotyping of various B cell subsets, including MZB and B1 B cells, to extend the data in Supplemental Fig 1H and I. Depending on the level of support, new immunization experiments to score Tfh and analyze a few of their functional molecules as part of a B cell paper may be feasible.

Addendum / update of Sept 2025: We added new data with more on MZB and B1 B cells, surface IgM, and on Tfh populations.

GC Response Analysis Upon Immunization: The GC response characterization should include additional data on the T cell compartment, specifically the presence and function of Tfh cells. In Fig. 1H, the distribution of the LZ appears strikingly different. However, the authors have not addressed this in the text. A more thorough characterization of centroblasts and centrocytes using CXCR4 and CD86 markers is needed.

The gating strategy used to characterize GC cells (GL7+CD95+ in IgD- cells) is suboptimal. A more robust analysis of GC cells should be performed in total B220+CD138- cells.

We first want to apologize the mislabeling of LZ and DZ in Fig 1H. The greenish-yellow colored region (GL7⁺ CD35⁺) indicate the DZ and the cyan-colored region (GL7⁺ CD35⁺) indicates the LZ. Addendum / update of Sept 2025: We corrected the mistake, and added new experimental data using the CD138 marker to exclude preplasmablasts.

As a technical note, we experienced high background noise with GL7 staining uniquely with PexRAP deficient (Dhrs7b^{f/f}; Rosa26-CreER^{T2}) mice (i.e., not WT control mice). The high background noise of GL7 staining was not observed in B cell specific KO of PexRAP (Dhrs7b^{f/f}; huCD20-CreER^{T2}). Two formal possibilities to account for this staining issue would be if either the expression of the GL7 epitope were repressed by PexRAP or the proper positioning of GL7⁺ cells in germinal center region were defective in PexRAPdeficient mice (e.g., due to an effect on positioning cues from cell types other than B cells). In a revised manuscript, we will fix the labeling error and further discuss the GL7 issue, while taking care not to be thought to conclude that there is a positioning problem or derepression of GL7 (an activation antigen on T cells as well as B cells).

While the gating strategy for an overall population of GC B cells is fairly standard even in the current literature, the question about using CD138 staining to exclude early plasmablasts (i.e., analyze B220⁺ CD138^{neg} vs B220⁺ CD138⁺) is interesting. In addition, some papers like to use GL7⁺ CD38^{neg} for GC B cells instead of GL7⁺ Fas (CD95)⁺, and we thank the reviewer for suggesting the analysis of centroblasts and centrocytes. For the revision, we will try to secure resources to revisit the immunizations and analyze them for these other facets of GC B cells (including CXCR4/CD86) and for their GL7⁺ CD38^{neg}, B220⁺ CD138⁻ and B220⁺ CD138⁺ cell populations.

We agree that comparison of the Rosa26-CreERT2 results to those with B cell-specific lossof-function raise a tantalizing possibility that Tfh cells also are influenced by PexRAP. Although the manuscript is focused on post-ontogenic B cell regulation in Ab responses, we hope to add a new immunization experiments that scores Tfh and analyzes a few of their functional molecules could be added to this B cell paper, depending on the ability to wheedle enough support / fiscal resources.

Addendum / update of Sept 2025: Within the tight time until lab closure, and limited \$\$, we were able to do experiments that further reinforced the GC B cell data - including stains for DZ vs LZ sub-subsetting - and analyzed Tfh cells. We were not able to explore changes in functional antigenic markers on the GC B or Tfh cells.

The authors claim that Dhrs7b supports the homeostatic maintenance of quiescent B cells in vivo and promotes effective proliferation. This conclusion is primarily based on experiments where CTV-labeled PexRAP-deficient B cells were adoptively transferred into μ MT mice (Fig. 2D-F). However, we recommend reviewing the flow plots of CTV in Fig. 2E, as they appear out of scale. More importantly, the low recovery of PexRAP-deficient B cells post-adoptive transfer weakens the robustness of the results and is insufficient to conclusively support the role of PexRAP in B cell proliferation in vivo.

In the revision, we will edit the text and try to adjust the digitized cytometry data to allow more dynamic range to the right side of the upper panels in Fig. 2E, and otherwise to improve the presentation of the in vivo CTV result. However, we feel impelled to push back respectfully on some of the concern raised here. First, it seems to gloss over the presentation of multiple facets of evidence. The conclusion about maintenance derives primarily from Fig. 2C, which shows a rapid, statistically significant decrease in B cell numbers (extending the finding of Fig. 1D, a more substantial decrease after a bit longer a period). As noted in the text, the rate of de novo B cell production does not suffice to explain the magnitude of the decrease.

In terms of proliferation, we will improve presentation of the Methods but the bottom line is that the recovery efficiency is not bad (comparing to prior published work) inasmuch as transferred B cells do not uniformly home to spleen. In a setting where BAFF is in ample supply in vivo, we transferred equal numbers of cells that were equally labeled with CTV and counted B cells. The CTV result might be affected by lower recovered B cell with PexRAP deficiency, generally, the frequencies of CTV^{low} divided population are not changed very much. However, it is precisely because of the pitfalls of in vivo analyses that we included complementary data with survival and proliferation in vitro. The proliferation was attenuated in PexRAP-deficient B cells in vitro; this evidence supports the conclusion that proliferation of PexRAP knockout B cells is reduced. It is likely that PexRAP deficient B cells also have defect in viability in vivo as we observed the reduced B cell number in PexRAP-deficient mice. As the reviewer noticed, the presence of a defect in cycling does, in the transfer experiments, limit the ability to interpret a lower yield of B cell population after adoptive transfer into μ MT recipient mice as evidence pertaining to death rates. We will edit the text of the revision with these points in mind.

In vitro stimulation experiments: These experiments need improvement. The authors have used anti-CD40 and BAFF for B cell stimulation; however, it would be beneficial to also include antiIgM in the stimulation cocktail. In Fig. 2G, CTV plots do not show clear defects in proliferation, yet the authors quantify the percentage of cells with more than three divisions. These plots should clearly display the gating strategy. Additionally, details about histogram normalization and potential defects in cell numbers are missing. A more in-depth analysis of apoptosis is also required to determine whether the observed defects are due to impaired proliferation or reduced survival.

As suggested by reviewer, testing additional forms of B cell activation can help explore the generality (or lack thereof) of findings. We plan to test anti-IgM stimulation together with anti-CD40 + BAFF as well as anti-IgM + TLR7/8, and add the data to a revised and final manuscript.

Addendum / update of Sept 2025: The revision includes results of new experiments in which anti-IgM was included in the stimulation cocktail, as well as further data on apoptosis and distinguishing impaired cycling / divisions from reduced survival .

With regards to Fig. 2G (and 2H), in the revised manuscript we will refine the presentation (add a demonstration of the gating, and explicate histogram normalization of FlowJo).

It is an interesting issue in bioscience, but in our presentation 'representative data' really are pretty representative, so a senior author is reminded of a comment Tak Mak made about a reduction (of proliferation, if memory serves) to 0.7 x control. [His point in a comment to referees at a symposium related that to a salary reduction by 30% :) A mathematical alternative is to point out that across four rounds of division for WT cells, a reduction to 0.7x efficiency at each cycle means about 1/4 as many progeny.]

We will try to edit the revision (Methods, Legends, Results, Discussion] to address better the points of the last two sentences of the comment, and improve the details that could assist in replication or comparisons (e.g., if someone develops a PexRAP inhibitor as potential therapeutic).

For the present, please note that the cell numbers at the end of the cultures are currently shown in Fig 2, panel I. Analogous culture results are shown in Fig 8, panels I, J, albeit with harvesting at day 5 instead of day 4. So, a difference of $\geq 3x$ needs to be explained. As noted above, a division efficiency reduced to 0.7x normal might account for such a decrease, but in practice the data of Fig. 2I show that the number of PexRAP-deficient B cells at day 4 is similar to the number plated before activation, and yet there has been a reasonable amount of divisions. So cell numbers in the culture of mutant B cells are constant because cycling is active but decreased and insufficient to allow increased numbers ("proliferation" in the true sense) as programmed death is increased. In line with this evidence, Fig 8G-H document higher death rates [i.e., frequencies of cleaved caspase3⁺ cell and Annexin V⁺ cells] of PexRAP-deficient B cells compared to controls. Thus, the in vitro data lead to the conclusion that both decreased division rates and increased death operate after this form of stimulation.

An inference is that this is the case in vivo as well - note that recoveries differed by $\sim 3x$ (Fig. 2D), and the decrease in divisions (presentation of which will be improved) was meaningful but of lesser magnitude (Fig. 2E, F).

Reviewer #2 (Public review):

Summary:

In this study, Cho et al. investigate the role of ether lipid biosynthesis in B cell biology, particularly focusing on GC B cell, by inducible deletion of PexRAP, an enzyme responsible for the synthesis of ether lipids.

Strengths:

Overall, the data are well-presented, the paper is well-written and provides valuable mechanistic insights into the importance of PexRAP enzyme in GC B cell proliferation.

We appreciate this positive response and agree with the overview and summary of the paper's approaches and strengths.

Weaknesses:

More detailed mechanisms of the impaired GC B cell proliferation by PexRAP deficiency remain to be further investigated. In the minor part, there are issues with the interpretation of the data which might cause confusion for the readers.

Issues about contributions of cell cycling and divisions on the one hand, and susceptibility to death on the other, were discussed above, amplifying on the current manuscript text. The aggregate data support a model in which both processes are impacted for mature B cells in general, and mechanistically the evidence and work focus on the increased ROS and modes of death. Although the data in Fig. 7 do provide evidence that GC B cells themselves are affected, we agree that resource limitations had militated against developing further evidence about cycling specifically for GC B cells. We will hope to be able to obtain sufficient data from some specific analysis of proliferation in vivo (e.g., Ki67 or BrdU) as well as ROS and death ex vivo when harvesting new samples from mice immunized to analyze GC B cells for CXCR4/CD86, CD38, CD138 as indicated by Reviewer 1. As suggested by Reviewer 2, we will further discuss the possible mechanism(s) by which proliferation of PexRAP-deficient B cells is impaired. We also will edit the text of a revision where to enhance clarity of data interpretation - at a

minimum, to be very clear that caution is warranted in assuming that GC B cells will exhibit the same mechanisms as cultures in vitro-stimulated B cells.

Addendum / update of Sept 2025: We were able to obtain results of intravital BrdU incorporation into GC B cells to measure cell cycling rates. The revised manuscript includes these results as well as other new data on apoptosis / survival, while deleting the data about CD138 populations whose interpretation was reasonably questioned by the referees.

Reviewer #1 (Recommendations for the authors):

We believe the evidence presented to support the role of PexRAP in protecting B cells from cell death and promoting B cell proliferation is not sufficiently robust and requires further validation in vivo. While the study demonstrates an increase in ether lipid content within the GC compartment, it also highlights a reduction in mature B cells in PexRAP-deficient mice under steady-state conditions. However, the IMS results (Fig. 3A) indicate that there are no significant differences in ether lipid content in the naïve B cell population. This discrepancy raises an intriguing point for discussion: why is PexRAP critical for B cell survival under steady-state conditions?

We thank the referee for all their care and input, and we agree that further intravital analyses could strengthen the work by providing more direct evidence of impairment of GC B cells in vivo. To revise and improve this manuscript before creation of a contribution of record, we performed new experiments to the limit of available funds and have both (i) added these new data and (ii) sharpened the presentation to correct what we believe to be one inaccurate point raised in the review.

(A) Specifically, we immunized mice with a B cell-specific depletion of PexRAP (Dhrs7b^{D/D-B} mice) and measured a variety of readouts of the GC B cells' physiology in vivo: proliferation by intravital incorporation of BrdU, ROS in the viable GC B cell gate, and their cell death by annexin V staining directly ex vivo. Consistent with the data with in vitro activated B cells, these analyses showed increased ROS (new - Fig. 7D) and higher frequencies of Annexin V⁺ 7AAD⁺ in GC B cells (GL7⁺ CD38⁻ B cell-gate) of immunized Dhrs7b^{D/D-B} mice compared with WT controls (huCD20-CreERT2^{+/-}, Dhrs7b^{+/-}) (new - Fig. 7E). Collectively, these results indicate that PexRAP aids (directly or indirectly) in controlling ROS in GC B cells and reduces B cell death, likely contributing to the substantially decreased overall GC B cell population. These new data are added to the revised manuscript in Figure 7.

Moreover, in each of two independent experiments (each comprising 3 vs 3 immunized mice), BrdU⁺ events among GL7⁺ CD38⁻ (GC B cell)-gated cells were reduced in the B cell-specific PexRAP knockouts compared with WT controls (new, Fig. 7F and Supplemental Fig 6E). This result on cell cycle rates in vivo is presented with caution in the revised manuscript text because the absolute labeling fractions were somewhat different in Expt 1 vs Expt 2. This situation affords a useful opportunity to comment on the culture of "P values" and statistical methods. It is intriguing to consider how many successful drugs are based on research published back when the standard was to interpret a result of this sort more definitively despite a merged "P value" that was not a full 2 SD different from the mean. In the optimistic spirit of the eLife model, it can be for the attentive reader to decide from the data (new, Fig. 7F and Supplemental Fig 6E) whether to interpret the BrdU results more strongly than what we state in the revised text.

(B) On the issue of whether or not the loss of PexRAP led to perturbations of the lipidome of B cells prior to activation, we have edited the manuscript to do a better job making this point more clear.

We point out to readers that in the resting, pre-activation state abnormalities were detected in naïve B cells, not just in activated and GC B cells. In brief, the IMS analysis and LC-MS-MS

analysis detected statistically significant differences in some, but not all, the ether phospholipids species in PexRAP deficient cells (some of which was in Supplemental Figure 2 of the original version).

With this appropriate and helpful concern having been raised, we realize that this important point merited inclusion in the main figures. We point specifically to a set of phosphatidyl choline ions shown in Fig. 3 (revised - panels A, B, D) of the revised manuscript (PC O-36:5; PC O-38:5; PC O-40:6 and -40:7).

For this ancillary record (because a discourse on the limitations of each analysis), we will note issues such as the presence of many non-B cells in each pixel of the IMS analyses (so that some or many "true positives" will fail to achieve a "significant difference") and for the naive B cells, differential rates of synthesis, turnover, and conversion (e.g., addition of another 2-carbon unit or saturation / desaturation of one side-chain). To the extent the concern reflects some surprise and perhaps skepticism that what seem relatively limited differences (many species appear unaffected, etc), we share in the sentiment. But the basic observation is that there are differences, and a reasonable connection between the altered lipid profile and evidence of effects on survival or proliferation (i.e., integration of survival and cell cycling / division).

Additionally, it would be valuable to evaluate the humoral response in a T-independent setting. This would clarify whether the role of PexRAP is restricted to GC B cells or extends to activated B cells in general.

We agree that this additional set of experiments would be nice and would extend work incrementally by testing the generality of the findings about Ab responses. The practical problem is that money and time ran out while testing important items that strengthen the evidence about GC B cells.

Finally, the manuscript would benefit from a thorough revision to improve its readability and clarity. Including more detailed descriptions of technical aspects, such as the specific stimuli and time points used in analyses, would greatly enhance the flow and comprehension of the study. Furthermore, the authors should review figure labeling to ensure consistency throughout the manuscript, and carefully cite the relevant references. For instance, S1PR2 CreERT2 mouse is established by Okada and Kurosaki (Shinnakasu et al, Nat. Immunol, 2016)

We appreciate this feedback and comment, inasmuch as both the clarity and scholarship matter greatly to us for a final item of record. For the revision, we have given our best shot to editing the text in the hopes of improved clarity, reduction of discrepancies (helpfully noted in the Minor Comments), and further detail-rich descriptions of procedures. We also edited the figure labeling to give a better consistency. While we note that the appropriate citation of Shinnakasu et al (2016) was ref. #69 of the original and remains as a citation, we have rechecked other referencing and try to use citations with the best relevant references.

Minor Comments: The labeling of plots in Fig. 2 should be standardized. For example, in Fig. 2C, D, and G, the same mouse strain is used, yet the Cre+ mouse is labeled differently in each plot.

We agree and have tried to tighten up these features in the panels noted as well as more generally (e.g., Fig. 4, 5, 6, 7, 9; consistency of huCD20-CreERT2 / hCD20CreERT2).

According to the text, the results shown in Fig. 1G and H correspond to a full KO (Dhrs7b^Δ/f; Rosa26-CreERT2 mice). However, Fig. 1H indicates that the bottom image corresponds to Dhrs7b^Δ/f, huCD20-CreERT2 mice (Dhrs7b^Δ/Δ -B).

We have corrected Fig. 1H to be labeled as Dhhrs7b^{Δ/Δ} (with the data on Dhhrs7b^{Δ/Δ-B} presented in Supplemental Figure 4A, which is correctly labeled). Thank you for picking up this error that crept in while using copy/paste in preparation of figure panels and failing to edit out the "-B"!

Similarly, the gating strategy for GC cells in the text mentions IgD⁻ cells, while the figure legend refers to total viable B cells. These discrepancies need clarification.

We believe we located and have corrected this issue in the revised manuscript.

Figures 3 and 4. The authors claim that B cell expression of PexRAP is required to achieve normal concentrations of ether phospholipids.

Suggestions for Improvement:

Lipid Metabolism Analysis: The analysis in Fig. 3 is generally convincing but could be strengthened by including an additional stimulation condition such as anti-IgM plus antiCD40. In Fig. 4C, the authors display results from the full KO model. It would be helpful to include quantitative graphs summarizing the parameters displayed in the images.

We have performed new experiments (anti-IgM + anti-CD40) and added the data to the revised manuscript (new - Supplemental Fig. 2H and Supplemental Fig 6, D & F). Conclusions based on the effects are not changed from the original.

As a semantic comment and point of scientific process, any interpretation ("claim") can - by definition - only be taken to apply to the conditions of the experiment. Nonetheless, it is inescapable that at least for some ether P-lipids of naive, resting B cells, and for substantially more in B cells activated under the conditions that we outline, B cell expression of PexRAP is required.

With regards to the constructive suggestion about a new series of lipidomic analyses, we agree that for activated B cells it would be nice and increase insight into the spectrum of conditions under which the PexRAP-deficient B cells had altered content of ether phospholipids. However, in light of the costs of metabolomic analyses and the lack of funds to support further experiments, and the accuracy of the point as stated, we prioritized the experiments that could fit within the severely limited budget.

[One can add that our results provide a premise for later work to analyze a time course after activation, and to perform isotopomer (SIRM) analyses with [13] C-labeled acetate or glucose, so as to understand activation-induced increases in the overall To revise the manuscript, we did however extrapolate from the point about adding BCR cross-linking to anti-CD40 as a variant form of activating the B cells for measurements of ROS, population growth, and rates of division (CTV partitioning). The results of these analyses, which align with and thereby strengthen the conclusions about these functional features from experiments with anti-CD40 but no anti-IgM, are added to Supplemental Fig 2H and Supplemental Fig 6D, F.

Figures 5, 6, and 7

The authors claim that Dhhrs7b in B cells shapes antibody affinity and quantity. They use two mouse models for this analysis: huCD20-CreERT2 and Dhhrs7b f/f; S1pr2-CreERT2 mice.

Suggestions for Improvement:

Adaptive immune response characterization: A more comprehensive characterization of the adaptive immune response is needed, ideally using the Dhhrs7b^{f/f}; S1pr2-CreERT2 model. This should include: Analysis of the GC response in B220+CD138⁻ cells. Class switch recombination analysis. A detailed characterization of centroblasts, centrocytes, and Tfh populations. Characterization of effector cells (plasma cells and memory cells).

Within the limits of time and money, we have performed new experiments prompted by this constructive set of suggestions.

Specifically, we analyzed the suggested read-outs in the huCD20-CreERT2, Dhhrs7b^{f/f} model after immunization, recognizing that it trades greater signal-noise for the fact that effects are due to a mix of the impact on B cells during clonal expansion before GC recruitment and activities within the GC. In brief, the results showed that

(a) the GC B cell population - defined as CD138^{neg} GL7⁺ CD38^{lo/neg} IgD^{neg} B cells - was about half as large for PexRAP-deficient B cells net of any early- or preplasmablasts (CD138⁺ events) (new - Fig 5G);

(b) the frequencies of pre- / early plasmablasts (CD138⁺ GL7⁺ CD38^{neg}) events (see new - Fig. 6H, I; also, new Supplemental Fig 5D) were so low as to make it unlikely that our data with the S1pr2-CreERT2 model (in Fig 7B, C) would be affected meaningfully by analysis of the CD138 levels;

(c) There was a modest decrease in centrocytes (LZ) but not centroblasts (DZ) (new - Fig 5H, I) - consistent with the immunohistochemical data of Supplemental Fig. 5A-C).

Because of time limitations (the "shelf life" of funds and the lab) and insufficient stock of the S1pr2-CreERT2, Dhhrs7b^{f/f} mice as well as those that would be needed as adoptive transfer recipients because of S1PR2 expression in (GC-)Tfh, the experiments were performed instead with the huCD20-CreERT2, Dhhrs7b^{f/f} model. We would also note that using this Cre transgene better harmonizes the centrocyte/centroblast and Tfh data with the existing data on these points in Supplemental Fig. 4.

(d) Of note, the analyses of Tfh and GC-Tfh phenotype cells using the huCD20-CreERT2 B cell type-specific inducible Cre system to inactivate Dhhrs7b (new - Supplemental Fig 1G-I; which, along with new - Supplemental Fig 5E) provide evidence of an abnormality that must stem from a function or functions of PexRAP in B cells, most likely GC B cells. Specifically, it is known that the GC-Tfh population proliferates and is supported by the GC B cells, and the results of B cell-specific deletion show substantial reductions in Tfh cells (both the GC-Tfh gating and the wider gate for plots of CXCR5/PD-1/ fluorescence of CD4 T cells

Timepoint Consistency: The NP response (Fig. 5) is analyzed four weeks postimmunization, whereas SRBC (Supp. Fig. 4) and Fig. 7 are analyzed one week or nine days post-immunization. The NP system analysis should be repeated at shorter timepoints to match the peak GC reaction.

This comment may stem from a misunderstanding. As diagrammed in Fig. 5A, the experiments involving the NP system were in fact measured at 7 d after a secondary (booster) immunization. That timing is approximately the peak period and harmonizes with the 7 d used for harvesting SRBC-immunized mice. So in fact the data with each system were obtained at a similar time point. Of course the NP experiments involved a second immunization so that many plasma cell and Ab responses derived from memory B cells generated by the primary immunization. However, the field at present is dominated by the view that the vast majority of the GC B cells after this second immunization (which historically we perform with alum adjuvant) are recruited from the naive rather than the

memory B cell pool. For the revised manuscript, we have taken care that the Methods, Legend, and Figure provide the information to readers, and expanded the statement of a rationale.

It may seem a technicality but under NIH regulations we are legally obligated to try to minimize mouse usage. It also behooves researchers to use funds wisely. In line with those imperatives, we used systems that would simultaneously allow analyses of GC B cells, identification of affinity maturation (which is minimal in our hands at a 7 d time point after primary NP-carrier immunization), and a switched repertoire (also minimal), and where with each immunogen the GC were scored at 7-9 d after immunization (9 d refers to the S1pr2-CreERT2 experiments). Apart from the end of funding, we feel that what little might be learned from performing a series of experiments that involve harvests 7 d after a primary immunization with NP-ovalbumin cannot well be justified.

In vitro plasma cell differentiation: Quantification is missing for plasma cell differentiation in vitro (Supp. Fig. 4). The stimulus used should also be specified in the figure legend. Given the use of anti-CD40, differentiation towards IgG1 plasma cells could provide additional insights.

As suggested by reviewer, we have added the results of quantifying the in vitro plasma cell differentiation in Supplemental Fig 6B. Also, we edited the Methods and Supplemental Figure Legend to give detailed information of in vitro stimulation.

Proliferation and apoptosis analysis: The observed defects in the humoral response should be correlated with proliferation and apoptosis analyses, including Ki67 and Caspase markers.

As suggested by the review, we have performed new experiment and analyzed the frequencies of cell death by annexin V staining, and elected to use intravital uptake of BrdU as a more direct measurement of S phase / cell cycling component of net proliferation. The new results are now displayed in Figure 5 and Supplemental Fig. 5.

Western blot confirmation: While the authors have demonstrated the absence of PexRAP protein in the huCD20-CreERT2 model, this has not been shown in GC B cells from the Dhhrs7b f/f; S1pr2-CreERT2 model. This confirmation is necessary to validate the efficiency of Dhhrs7b deletion.

We were unable to do this for technical reasons expanded on below. For the revision, we have edited in a bit of text more explicitly to alert readers to the potential impact of counterselection on interpretation of the findings with GC B cells. Before entering the GC, B cells have undergone many divisions, so if there were major pre-GC counterselection, in all likelihood the GC B cells would PexRAP-sufficient. To recap from the original manuscript and the new data we have added, IMS shows altered lipid profiles in the GC B cells and the literature indicates that the lipids are short-lived, requiring de novo resynthesis. The BrdU, ROS, and annexin V data show that GC B cells are abnormal. Accordingly, abnormal GC B cells represent the parsimonious or straightforward interpretation of the new results with GC-Tfh cell prevalence.

While we take these findings together to suggest that counterselection (i.e., a Western result showing normal levels of PexRAP in the GC B cells) seems unlikely, it is formally possible and would mean that the in situ defects of GC B cells arose due to environmental influences of the PexRAP-deficient B cells during the developmental history of the WT B cells observed in the GC.

Having noted all that, we understand that concerns about counter-selection are an issue if a reader accepts the data showing that mutant (PexRAP-deficient) B cells tend to proliferate less and die more readily. Indeed, one can speculate that were we also to perform competition experiments in which the Ighb, Cd45.2 B cells (WT or Dhhrs7b D/D) are mixed with equal numbers of Igha, Cd45.1 competitors, the differences would become much greater. With this in mind, Western blotting of flow-purified GC B cells might give a sense of how much counter-selection has occurred.

That said, the Westerns need at least 2.5×10^6 B cells (those in the manuscript used five million, 5×10^6) and would need replication. Taken together with the observation that ~200,000 GC B cells (on average) were measured in each B cell-specific knockout mouse after immunization (Fig. 1, Fig 5) and taking into account yields from sorting, each Western would require some 20-25 tamoxifen-injected ___-CreERT2 , Dhhrs7b f/f mice, and about half again that number as controls. The expiry of funds prohibited the time and costs of generating that many mice (>70) and flow-purified GC B cells.

Figure 8

The authors claim that Dhhrs7b contributes to the modulation of ROS, impacting B cell proliferation.

Suggestions for Improvement:

GC ROS Analysis: The in vitro ROS analysis should be complemented by characterizing ROS and lipid peroxidation in the GC response using the Dhhrs7b f/f; S1pr2-CreERT2 model. Flow cytometry staining with H2DCFDA, MitoSOX, Caspase-3, and Annexin V would allow assessment of ROS levels and cell death in GC B cells.

While subject to some of the same practical limits noted above, we have performed new experiments in line with this helpful input of the reviewer, and added the helpful new data to the revised manuscript. Specifically, in addition to the BrdU and phenotyping analyses after immunization of huCD20-CreER^{T2}, Dhhrs7b^{f/f} mice, DCFDA (ROS), MitoSox, and annexin V signals were measured for GC B cells. Although the mitoSox signals did not significantly differ for PexRAP-deficient GCB, the ROS and annexin V signals were substantially increased. We added the new data to Figure 5 and Supplemental Figure 5. Together with the decreased in vivo BrdU incorporation in GC B cells from Dhhrs7b^{D/D-B} mice, these results are consistent with and support our hypothesis that PexRAP regulates B cell population growth and GC physiology in part by regulating ROS detoxification, survival and proliferation of B cells.

Quantification is missing in Fig. 8E, and Fig. 8F should use clearer symbols for better readability.

We added quantification for Fig 8E in Supplemental Fig 6E, and edited the symbols in Fig 8F for better readability.

Figure 9

The authors claim that Dhhrs7b in B cells affects oxidative metabolism and ER mass. The results in this section are well-performed and convincing.

Suggestion for Improvement:

Based on the results, the discussion should elaborate on the potential role of lipids in antigen presentation, considering their impact on mitochondria and ER function.

We very much appreciate the praise of the tantalizing findings about oxidative metabolism and ER mass, and will accept the encouragement that we add (prudently) to the Discussion section to make note of the points mentioned by the Reviewer, particularly now that (with their encouragement) we have the evidence that B cell-specific loss of PexRAP (with the huCD20-CreERT2 deletion prior to immunization) resulted in decreased (GC-)Tfh and somewhat lower GC B cell proliferation.

Reviewer #2 (Recommendations for the authors):

The authors should investigate whether PexRAP-deficient GC B cells exhibit increased mitochondrial ROS and cell death ex vivo, as observed in in vitro cultured B cells.

We very much appreciate the work of the referee and their input. We addressed this helpful recommendation, in essence aligned with points from Reviewer 1, via new experiments (until the money ran out) and addition of data to the manuscript. To recap briefly, we found increased ROS in GC B cells along with higher fractions of annexin V positive cells; intriguingly, increased mtROS (MitoSox signal) was not detected, which contrasts with the results in activated B cells in vitro in a small way. To keep the text focused and not stray too far outside the foundation supported by data, this point may align with papers that provide evidence of differences between pre-GC and GC B cells (for instance with lack of Tfam or LDHA in B cells).

It remains unclear whether the impaired proliferation of PexRAP-deficient B cells is primarily due to increased cell death. Although NAC treatment partially rescued the phenotype of reduced PexRAP-deficient B cell number, it did not restore them to control levels. Analysis of the proliferation capacity of PexRAP-deficient B cells following NAC treatment could provide more insight into the cause of impaired proliferation.

To add to the data permitting an assessment of this issue, we performed new experiments in which B cells were activated (BCR and CD40 cross-linking), cultured, and both the change in population and the CTV partitioning were measured in the presence or absence of NAC. The results, added to the revision as Supplemental Fig 6FH, show that although NAC improved cell numbers for PexRAP-deficient cells relative to controls, this compound did not increase divisions at all. We infer that the more powerful effect of this lipid synthesis enzyme is to promote survival rather than division capacity.

Primary antibody responses were assessed at only one time point (day 20). It would be valuable to examine the kinetics of antibody response at multiple time points (0, 1w, 2w, 3w, for example) to better understand the temporal impact of PexRAP on antibody production.

We thank the reviewer for this suggestion. While it may be that the kinetic measurement of Ag-specific antibody level across multiple time points would provide an additional mechanistic clue into the of impact PexRAP on antibody production, the end of sponsored funding and imminent lab closure precluded performing such experiments.

CD138+ cell population includes both GC-experienced and GC-independent plasma cells (Fig. 7). Enumeration of plasmablasts, which likely consists of both PexRAP-deleted and undeleted cells (Fig. 7D and E), may mislead the readers such that PexRAP is dispensable for plasmablast generation. I would suggest removing these data and instead examining the number of plasmablasts in the experimental setting of Fig. 4A (huCD20-CreERT2-mediated deletion) to address whether PexRAP-deficiency affects plasmablast generation.

We have eliminated the figure panels in question, since it is accurate that in the absence of a time-stamping or marking approach we have a limited ability to distinguish plasma cells that arose prior to inactivation of the *Dhrs7b* gene in B cells. In addition, we performed new experiments that were used to analyze the "early plasmablast" phenotype and added those data to the revision (Supplemental Fig 5D).

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